Unit : 1 Introduction

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(1) What is computer network ? Describe the various types of computer n/w.

→ Computer network :-

A computer network is a system in which multiple computer are connected to each other to share information and resources.

→ Types of computer network :-

(1). LAN
(Local area network)

- easy to design and troubleshoot.
- In LAN, all the machines are connected to a single cable.
- Different types of topologies such as Bus, ring, star, tree are used.
- The data transfer speed is 10 Gb/s
- They exist in limited geographical

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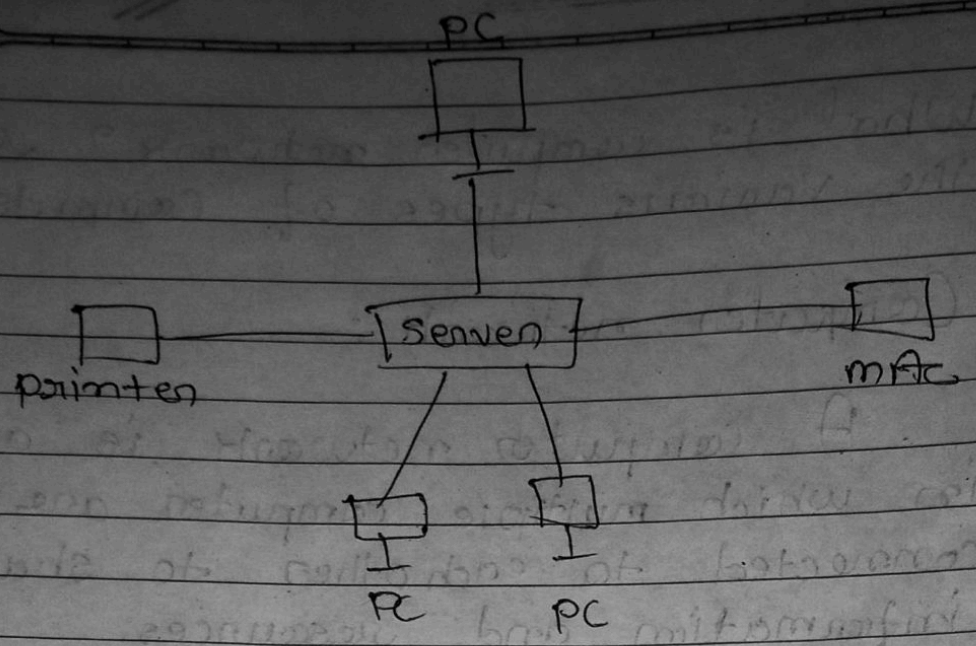
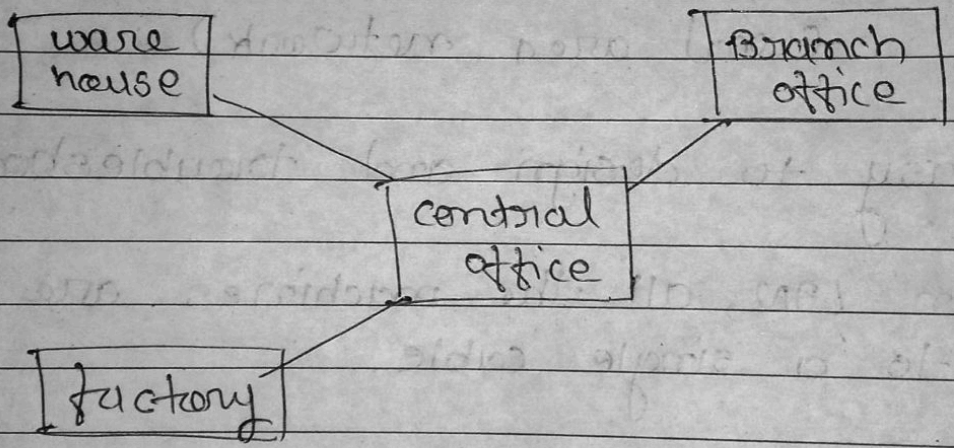


(2) MAN
(Metropolitan Area Network)



- MAN connects two or more LANs.
- It covers a city.
- The best example of MAN is the cable television network available in many cities.

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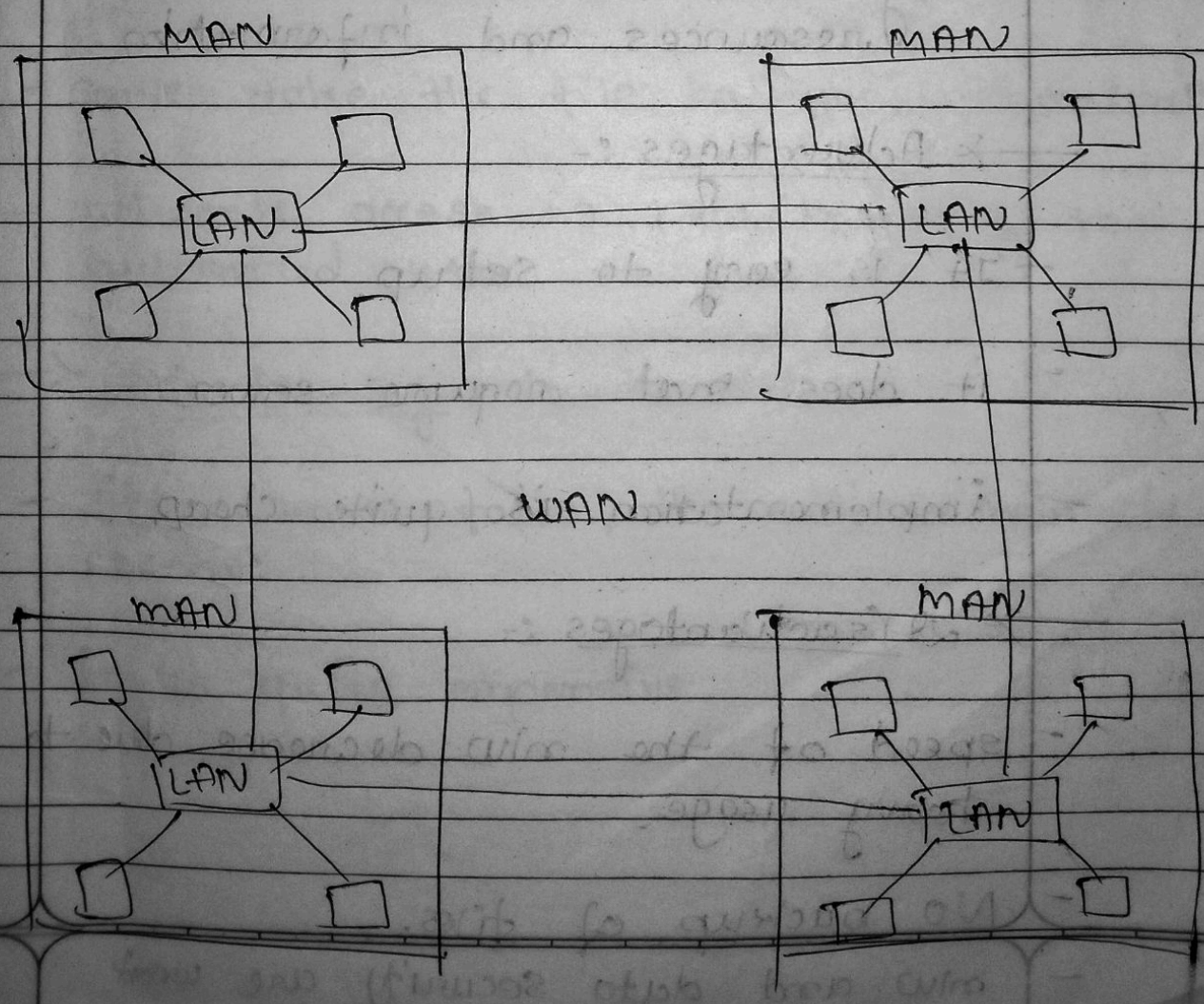
(3). WAN
(Wide Area Network)

- it cover a large geographical area often a country.
- It may be located entirely within a state or country or around the world.
- The communication between different users of WAN is established using leased telephone lines or satellite and similar channels.



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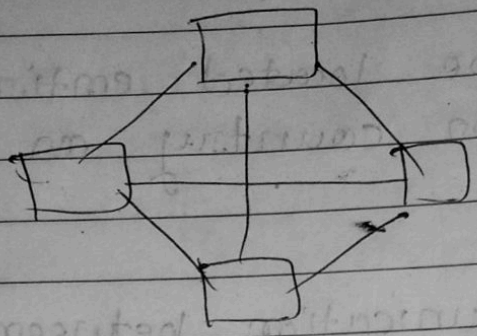
(2) What is peer to peer and client server model ? Mention advantages and disadvantages of both models.

⟹ peer to peer model :-



- work as client & server both.
- Group of computer is connected together so that user can share resources and information.

→ Advantages :-

- It is easy to set up.
- it does not require server.
- implementation is quite cheap.

→ Disadvantages :-

- speed of the nlw decrease due to heavy usage.
- No backup of files.
- nlw and data security are weak

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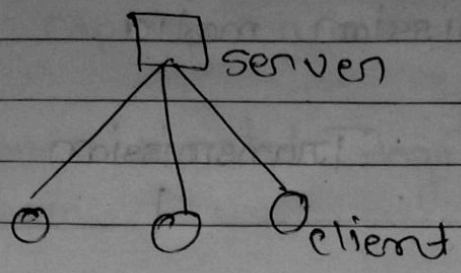
Client/server network model :-



- It is a system where one or more computers called clients to connect to a central computer named as a server to share or use resources.

Advantages :-

- The server holds the shared files.
- Server take the file backups automatically.
- network access is provide only to authorized user.

Disadvantages :-

- if a server fails, the entire network crashes.
- it is quite expensive.

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{3} Discribe the different types of transmission media.

→ Transmission media

↓ Guided media
1. Twisted pair cable
2. Coaxial cable
3. fiber optic cable

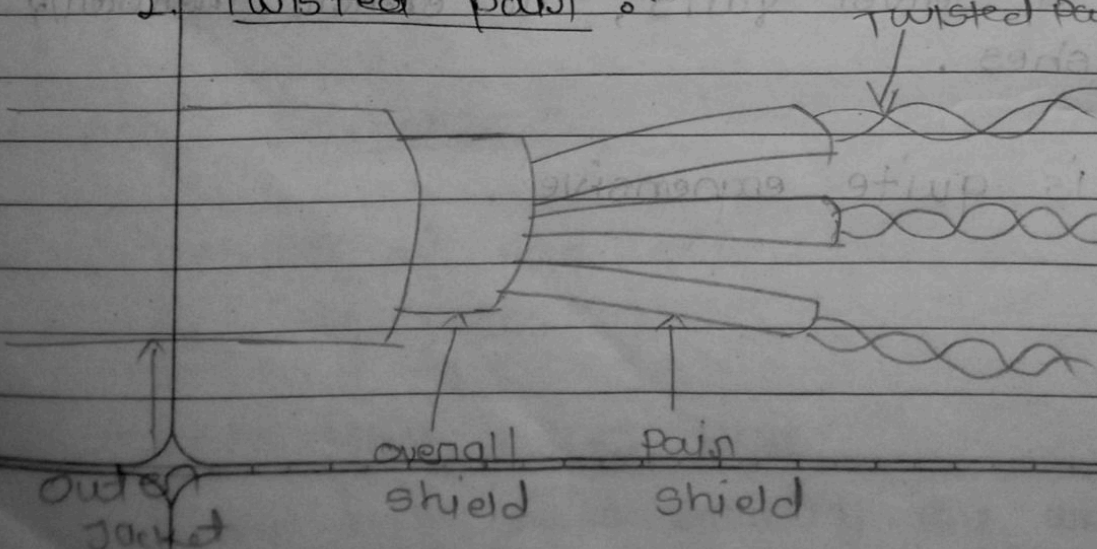
↓ unguided media
1. Radio wave
2. Micro wave
3. Infrared wave

* Guided media :-

Guided media are those that provide a channel from one device to another.

1. Twisted pair
2. Coaxial
3. fiber optic

1. Twisted pair :-



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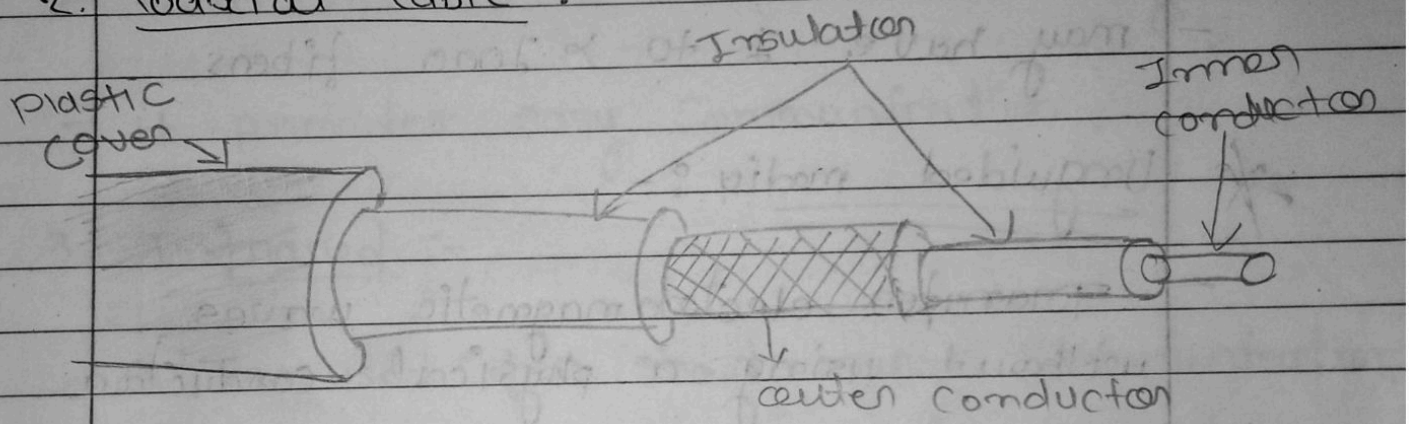
- A twisted pair consists of two insulated copper wires.
- The wire are twisted together in a helical form.
- it also contain color coded plastic insulation.

2. Coaxial cable :-



- it has better shielding than twisted pairs,
- it use for longer distances at higher speed.
- Coaxial cables used for telephone, for long distance lines.
- also called coax.
- it type of shielded and insulated copper cable.

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3. Fiber optic :-

- is made of glass or plastic.
- it transmits signals in light form.
- it is the fastest cable.
- The light is lasser form.
- may have 1 to > 1000 fibers

* Unguided media :-

- Transfer electromagnetic waves without using a physical conduction.
- also known as wireless Communication.

1. Radio wave
2. Micro wave
3. Infrared

1. Radio wave :-

- easy to generate.
- Can travel long distances.
- Radio waves also are omnidirectional meaning that they travel in all direction.

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- It is frequency dependant.

2. Micro wave :-

- Travels in straight light.
- Communication over oceans can be achieved by using microwave transmission.
- transmission is cheaper than using cables.
- it provides easy communication.

3. Infrared :-

- it is used for short range communication.
- Eg. TV, VCRs, cell phone.
- wireless technology.
- provides better security.
- High bandwidth.

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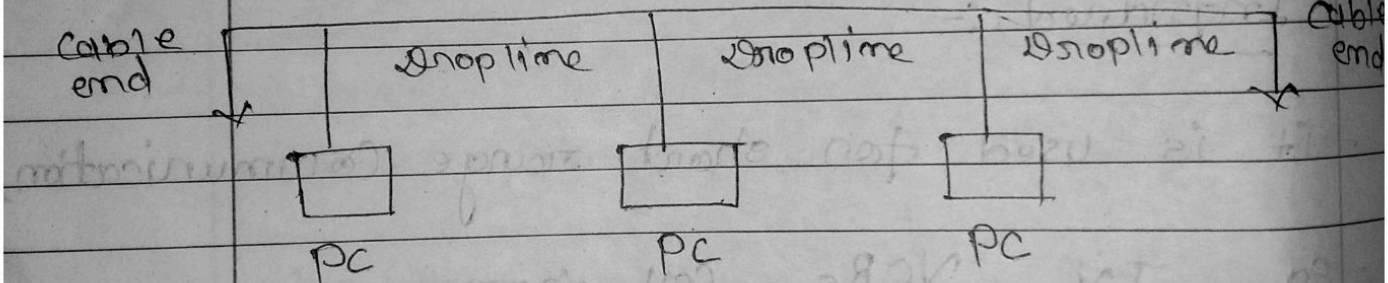
(4) Describe the different network topologies with diagram. mention their advantages and disadvantages.

→ 1. Bus
2. Ring
3. Star
4. Mesh
5. Tree
6. Hybrid

(1) Bus topology :-

- every computer and network device is connected to a single cable.



→ Advantages :-

- Cost is cheapen.
- Used in small network.
- easy to understand.
- cable required is least compared to other network network.

→ Disadvantages :-

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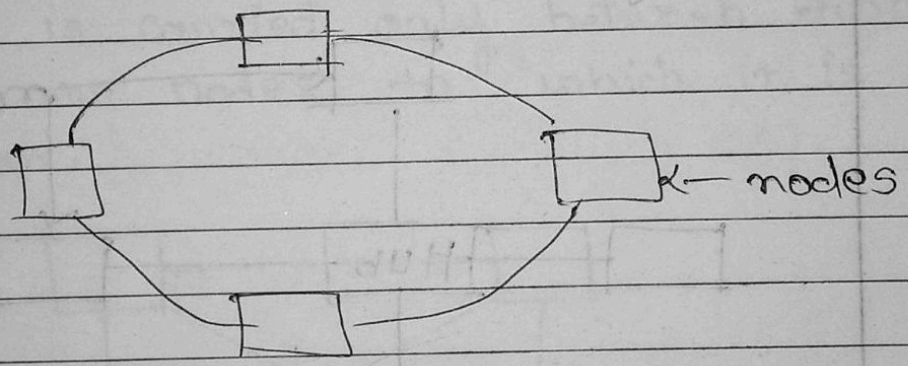
- Cable fails then the whole nlw fails.
- Cable has limited length.
- If network traffice is heavy or nodes are more the performance of the network decreases.

② Ring topology :-

- it forms a ring as each computer is connected to another computer with the last one connected to the first.



→ Advantages :-

- cheap to install and expand.
- only the node having token then transmit data.

→ Disadvantages :-

- Troubleshooting is difficult.

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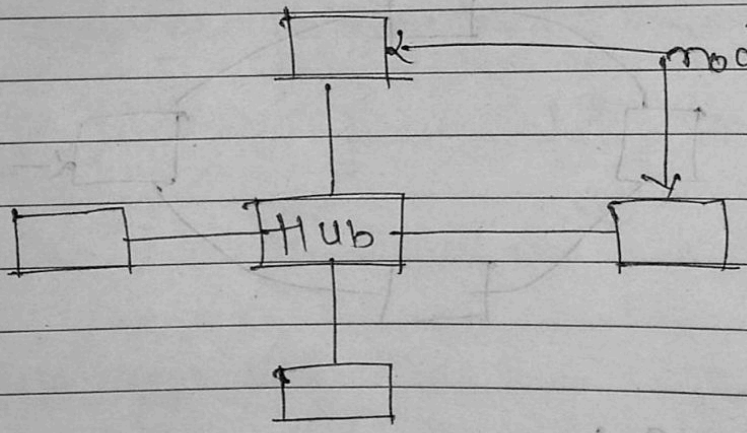
- Adding or deleting the Computers disturbs the network activity.
- failure of one computer disturbs the whole network.

③. Star topology :-

- every node have it's own Connection line.
- In this topology, all the Computers are connected to a single hub through a cable.



→ Advantages :-

- fast performance.
- Hub can be upgraded easily.
- Easy to troubleshoot.
- Easy to setup and modify.

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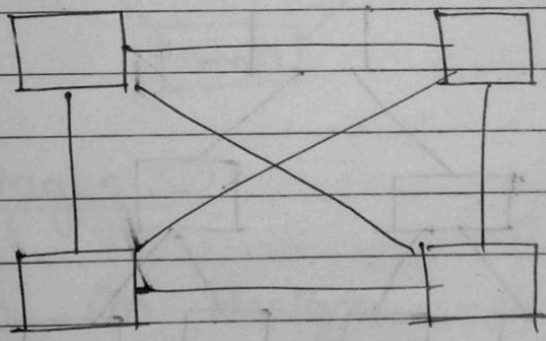
→ Disadvantages :-

- Cost of installation is high
- Expensive to use.
- performance is based on the hub.

④. Mesh topology :-

- It is point to point connection to other nodes or devices.
- Traffic is carried only between two devices or nodes to which it is connected.



→ Advantages :-

- each connection can carry it's own data load.
- provide security and privacy.
- A fault is diagnosed easily.

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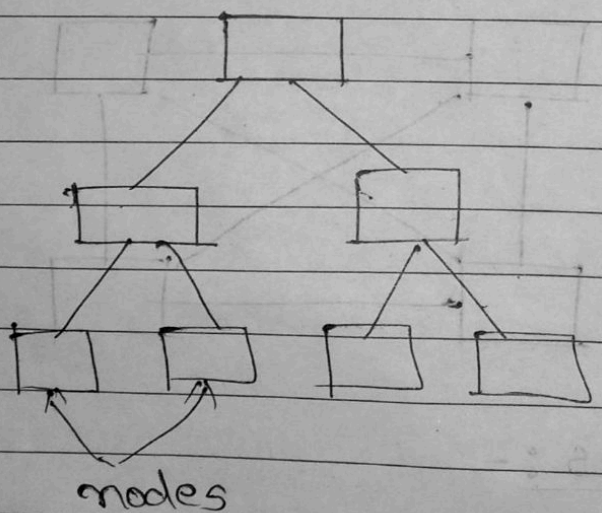
→ Disadvantages :-

- Installation and configuration are difficult.
- Cabling cost is more.
- Bulk wiring is required.

⑤ Tree topology :-

- It has a root node and all other nodes are connected to it forming a hierarchy.
- also called hierarchical topology.



→ Advantages :-

- Extension of star and bus.
- Expansion of nodes is possible and easy.

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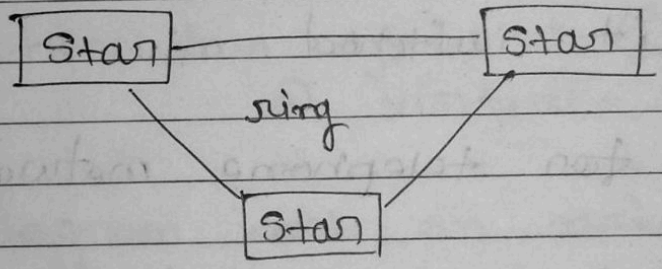
- error detection is easily done.

→ Disadvantages :-

- Heavily cable
- Costly
- Central hub fails then network fails.

(6) Hybrid topology :-

- A network structure whose design contains more than one topology is said to be hybrid topology.



→ Advantages :-

- Complex in design.
- Costly.

(5) Describe all the classification of a switching network.

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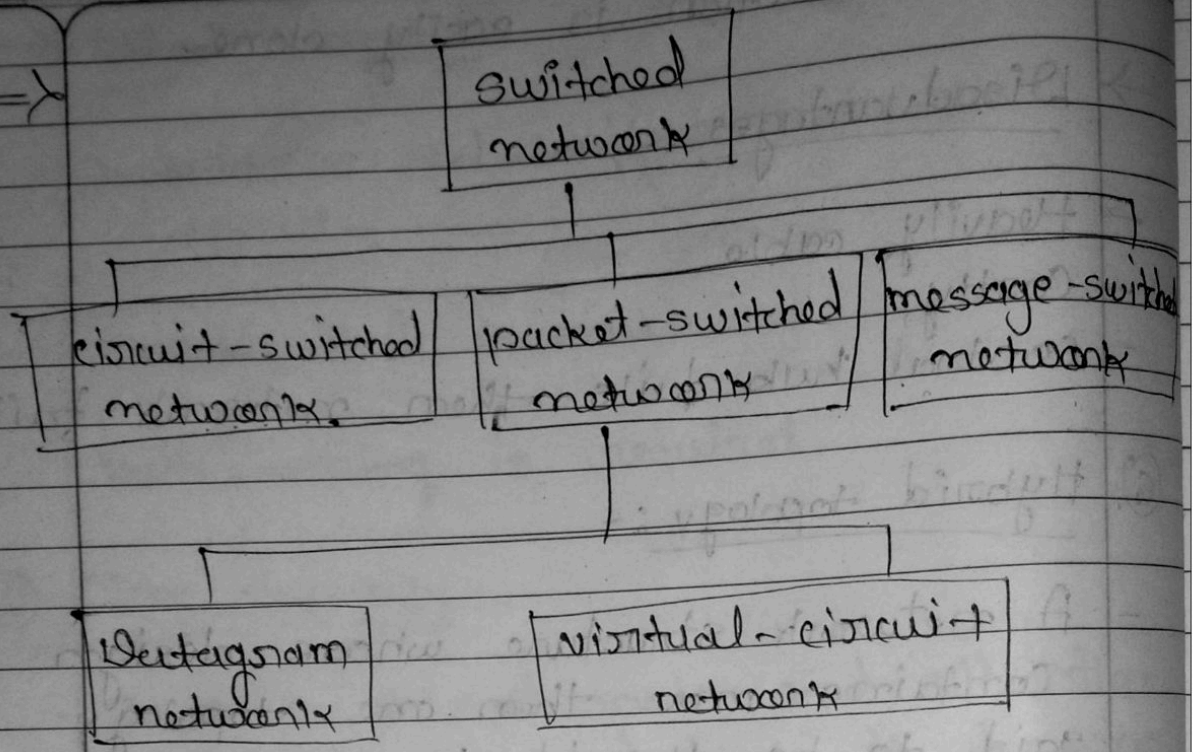

→ Circuit switched ntw :-

- used for telephone networks.
- it work on physical layer.
- Circuit switched network using dedicated path.
- Communication are in Contiguous flow. for eg. we say "Hello How are you."

then it receive "Hello How are you."

- Data are properly flow.

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- efficiency less.
- delays less.
- it is not use for message, email, computer networks. it's only good for telephone network.

→ packet - switched n/w :-

- Data are divided into small packets, then it transmit.
- it has two types :- ①. Datagram n/w
  ②. virtual - circuit n/w

- Datagram work on network layer.
- virtual - circuit work on Data link layer.
- it can store data and then forward the data.
- that's why it's more efficient.
- More delay in packet switching.
- it used pipeling concept.
- Packets are send paraller.

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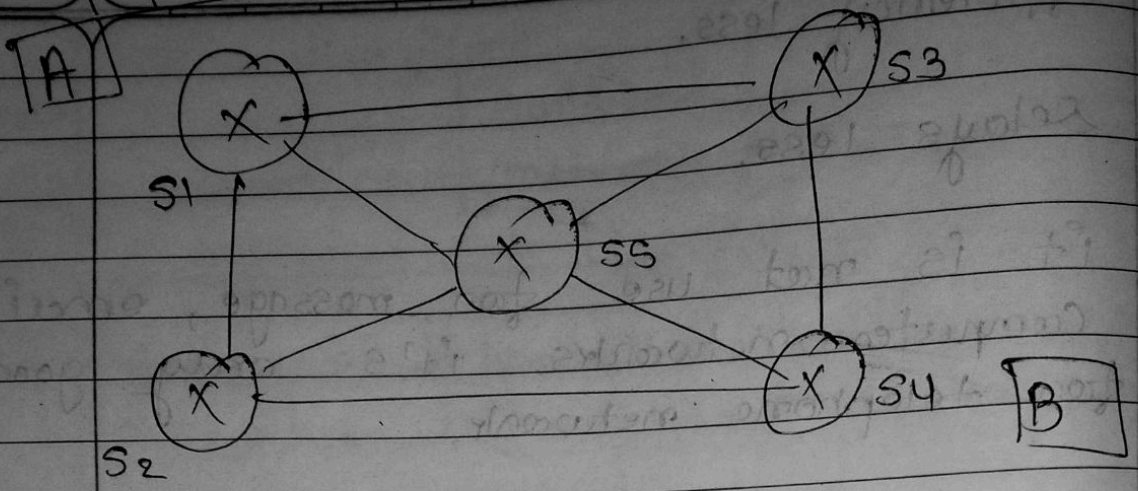
4|3|2|1 ← packets



②. Datagram

- it is connectionless.
- each packet treated independently.
- it use no reservation.
- packets are follow different path. using routing table. (out of order.)
- packet lost is high.
- it is used in internet.
- cost is low.
- More efficiency.
- More delay.

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② Virtual circuit

- it is connection - oriented.
- it use reservation.
- packets are follow same order path.
- packet lost is less.
- it is used in ATM.
  (Asynchronous transfer mode.)
- Cost is high.
- less delay.

→ Message switching :-

- this technique is somewhere middle of circuit switching & packet switching.
- whole message is treated as a data unit.
- message are stored in hop.
- in this hop - ~~by~~ - hop delivery.

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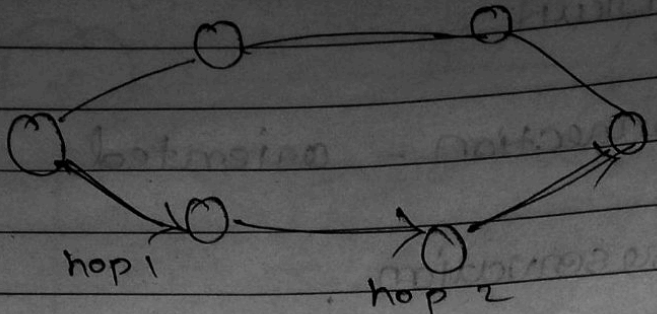

- Send data A to D using hop.

(6) OSI Model.

- physical layer
- Datalink layer
- Network layer
- Transport layer
- Session layer
- Presentation layer
- Application layer

① physical layer :-

- responsible for the actual physical connection between the device.
- Contain information byte (0, 1)

② Datalink layer :-

- Contain data in frames forms

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- frame error checking.
- provide error free transfer data, from one node to another.

③. Network layer :-

- Transmission of data from one host to other located in different network.
- routes frames among networks.
- Convert logical address in to physical address.

④. Transport layer :-

- Data in segment form.
- responsible for the end to end delivery of complete message in sequence without loss or duplication.

⑤. Session layer :-

- allow two app^{n} processes on different machine to establish use and terminate a connection, called a session.

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⑥ Presentation layer :-

- formates the data to be presented to the appⁿ layer.
- Data encrypt for security purpose.
- reduse the no. of bit & transmit on the network.

⑦ Application layer :-

- it is highest level.
- providing services directly for the appⁿ process.
- it allows user to access, retrieve and manage files in a remote computer.

Tcp/Ip Model

physical layer
Datalink layer
network layer
Transport layer
Application layer

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② Application layer :-

- Top most layer in TCP/IP.
- it handle high level protocol.
- allow user to intract with the application.

③ Transport layer :-

- provide end to end data transfer.
- responsible for flow control and connection of data.
- UDP, TCP protocal are used.

④ Network layer :-

- Data are available in packet form.
- it called ip datagram.
- in ipdatagram contain source and destination address.
- it provide routing function.

(6.) Network interface layer :-

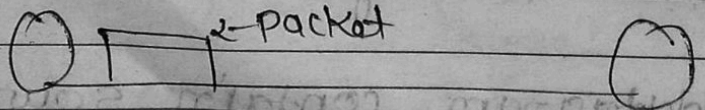
- it is combination of physical and Datalink.
- define how the data sent physicaly through the network.
- responsible for transmission for the data between two device on same network.

(7) Write short note on various delays. what is loss and throughput in a packet switching network?

⟹ Delay :-

1. Transmission delay
2. propagation delay
3. queuing delay
4. processing delay.

(1) Transmission delay :-



The time taken to transmit a packet from the host to the transmission medium is called transmission delay.

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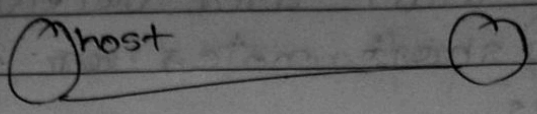
② Propagation delay :-



bit push in link, the time taken from one end to another end is called propagation delay.

③ Quing delay :-

packet is waiting in queue is called queing delay.

④ processing delay :-

The packet will be taken from processing which is called processing delay.

⟹ packet loss :-

It is the failur of one or more transmitted packets to arrive at their destination.

⟹ Throughput :-

It is the rate of successful message delivery over a communication channel.

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Unit :- 1

Q6) Disadvantages and Advantages of Computer networks.

⟹ Advantages :-

- file sharing.
- Resource sharing.
- Better connectivity and communications.
- Internet access
- Entertainment
- Inexpensive system
- Flexible access
- Instant and Multiple access

⟹ Disadvantages :-

- Lack of data security and privacy
- presence of Computer viruses and malware
- Lack of Independence
- Lack of Robustness
- Need an efficient handler.

Q9. Application of Computer network.

- financial services
- Email Services
- Mobile applications
- Directory services
- Teleconferencing

Q10. Internet.

- is a type of world-wide computer network.
- it's spread across the world.
- it connects hundreds of millions of computing devices throught the world.
- it established the large network.
- when two computers are ~~comp~~ connected over the internet they can send and receive all kind of info.

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Q11. Techniques used in data communications to transfer data.

→ 1. Connection-oriented Method.
2. Connectionless Method.

1. Connection-oriented :-

- it is just like voice phone call.
- There are two types :-

① without virtual circuits

- This is what Tcp does. on the internet.
- only two computers are. connection to establish with endpoints.
- routers and link have no information of connection.

② with virtual circuits

- this is not used on internet, but use in other type of network.
- The routers within the network route all packets in one connection over the same route.

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2 Connectionless :-

- is just packet switching where no call establishment and release occur.
- A message is broken into packets, and each packet is transferred separately.
- Connectionless service is typically provided by the UDP.

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(8) Mention the protocols used for sending mail and receiving mail. write short notes on these protocols.

⇒
1. pop 3
2. IMAP
3. SMTP

(1). pop 3 :-

- (post office protocol)
- it is a simple mail access protocol.
- client opens a Tcp connection to the mail server.
- Three phase in Tcp :- authorization
  Transaction
  update

(2) IMAP :-

- (Internet message Access protocol)
- in this using pop3 download the message to the local machine and store whenever he wants to.
- a reciver can delete message, move message among folder.

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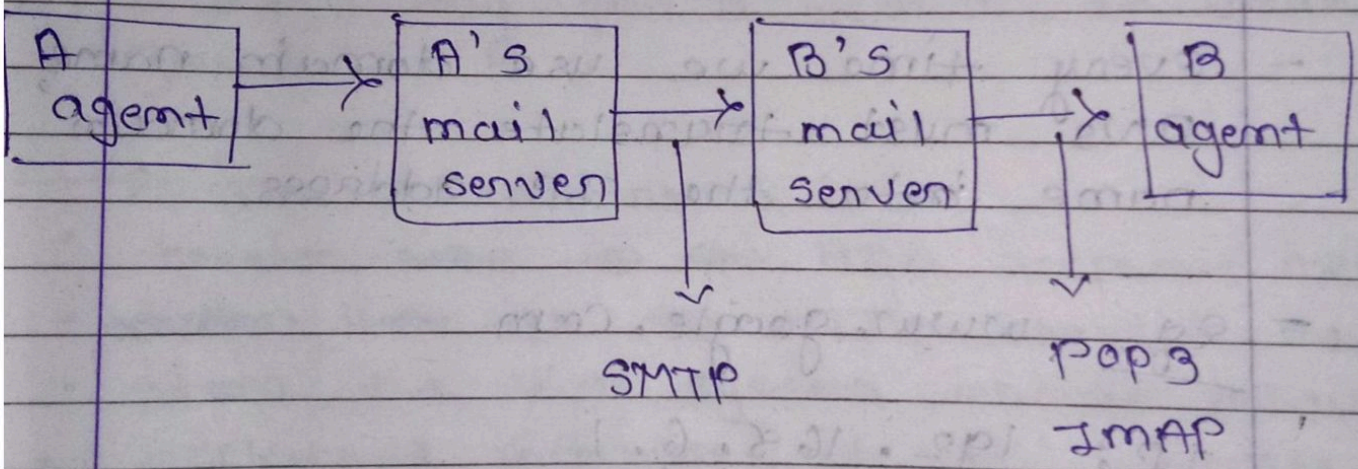
③ SMTP :-

- (simple mail Transfer protocol.)
- it transfer message from server to receiver.
- it convert the body of mail mail messages to simple 7-bit ASCII.



(9). Describe DNS. Mention how distributed database work.

→ DNS :-

(Domain name system.)

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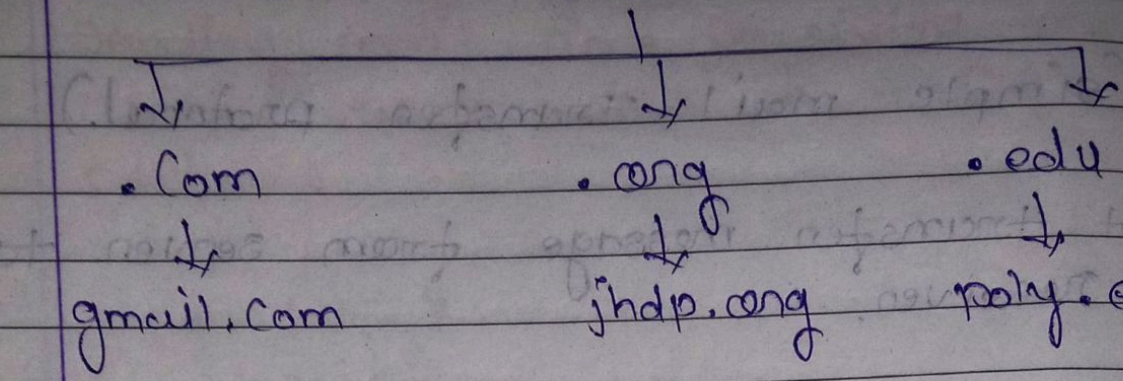

- Translate domain name into IP address.
- Domain name is alphabetic, that user can easy to remember.
- every time we use domain name, DNS must translate the domain name into the IP address.
- Eg. www.google.com
  IP: 192.168.6.1

→ client first contact Root server which return IP address of Top level domain

↓

→ then contact with TLC and return authorized server ip address.

↓

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→ finally client connect authenticated server.

↓

→ return ip of host name.

(10) How Cookies technology works?

⇒ Cookies are small text file, that created by website, and stored in user computer.

- it is a way for a website to remember you.

→ four components :-

- ~~boolean time in the~~ HTTP response message
- ~~boolean time in the~~ HTTP request message
- store the user system, manage browser
- back-end database at the website

→ working :-

- User access amazon.com first time.
- When request come into amazon server create a unique id number and create a entry in backend database.
- amazon web server respone to user web browser.

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- that include HTTP response, set cookie, header which contain the id number

→ eg - header line set cookie. 1617
- then user browser receive message.
- it see the set cookie, header.
- then browser append cookie file and manage.
- after some time user visit the same website.
- user continue the browser amazon site, because browser already include the cookie file.

→ use of cookie.

① authorization ③ recommendations
② shopping carts ④ user session state (webemail)

11) Mention the difference.

HTTP	SMTP
- It is a ~~push~~ pull protocol.	- it is a push protocol
- HTTP can use both persistent and non-persistent connection.	- SMTP uses a persistent connection.
- By default, HTTP uses port 80.	- 7-bit ASCII.
- It doesn't have any restriction to send message.	- It have restriction to send message.

POP3	IMAP
- To read the mail it has to be downloaded first.	- The mail content can be checked partially before downloading.
- The user cannot organize the mail to the mailbox of the mail server.	- The user can organize the mails on the server.
- it has limited function.	- It has more powerful, more complex & has more features over POP3

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Connection - oriented	Connection less
- It is related to the telephone system.	is related to the postal system.
- connection - oriented service is necessary	Connection - less service is not compulsory.
- feasible	not feasible
- congestion is not possible.	congestion is possible.
packets follow the same route.	packets do not follow the same route.

(12) Describe the HTTP protocol with diagram. Mention the format of HTTP request and HTTP response.

⟹ HTTP :-

(Hyper text transfer protocol)

which defines how web clients request webpages from web server and how servers transfer web pages to clients.

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- HTTP client establish Tcp connection with server.

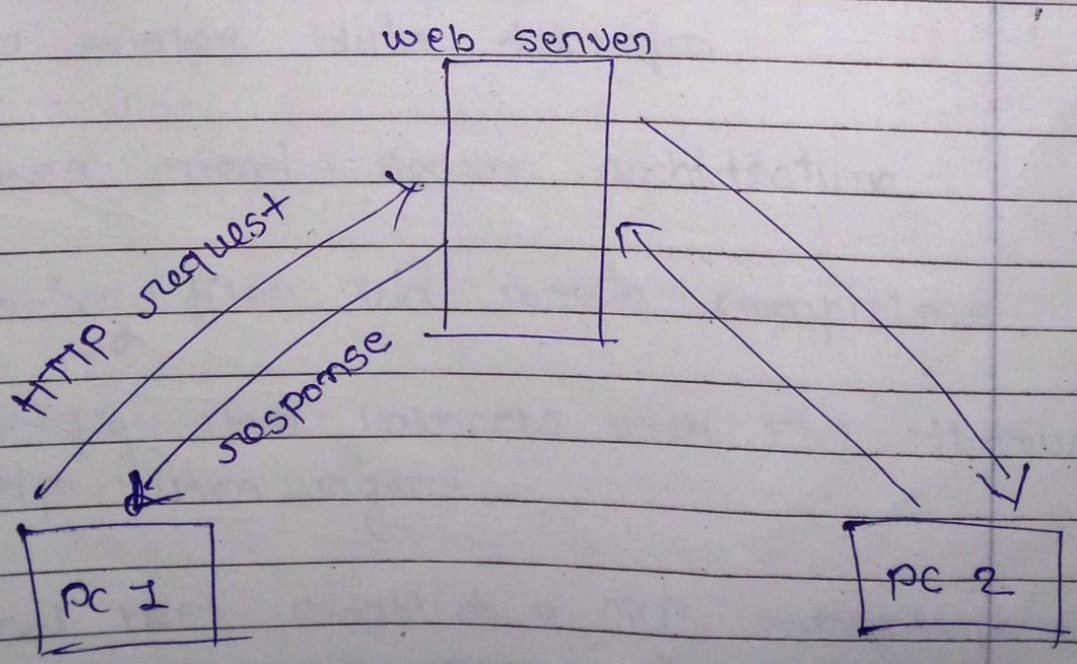
→ Two connection type :-

① Non-persistent HTTP

- for multiple objects it required multiple connections.
- it used one request and one response.

② persistent HTTP

- for multiple objects it required one connection.
- with persistent connections, the performance is improved by 20%.



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Message formate

1. Request format
2. response formate

1. Request formate :-

There are three parts :

1. Request line
2. header line
3. carriage line

2. Response formate :-

There are three parts :

1. Status line
2. header line
3. Data

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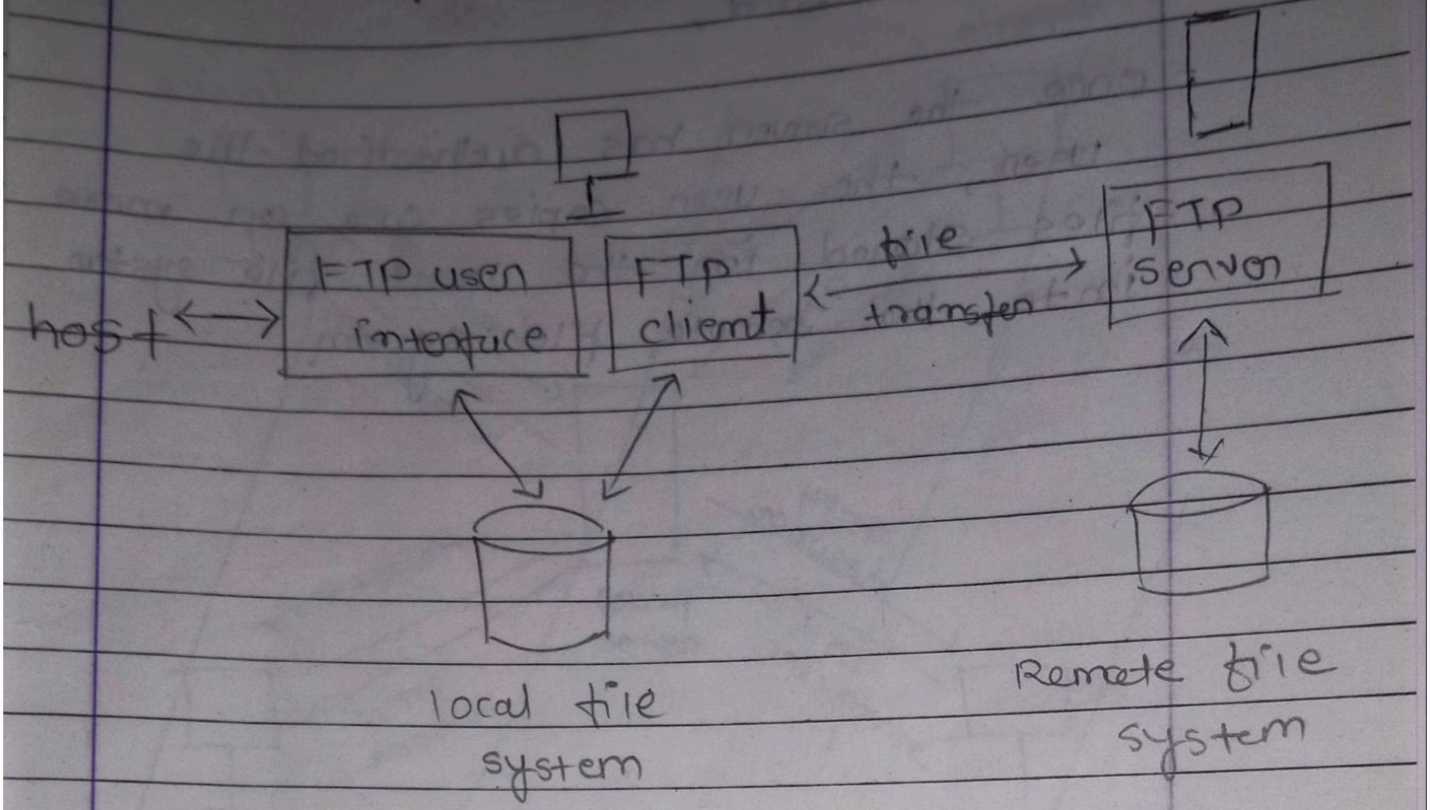
13) FTP (file transfer protocol).



- file transfer protocol is the commonly used protocol for exchanging files over the network on internet.
- FTP use the internet's TCP/IP protocols to enable data transfer.
- uses client - server architecture.
- sharing files via remote computers.
- in fig. user interacts with FTP through FTP user agent.
- local host establish a TCP connection with the FTP server process in the remote host.

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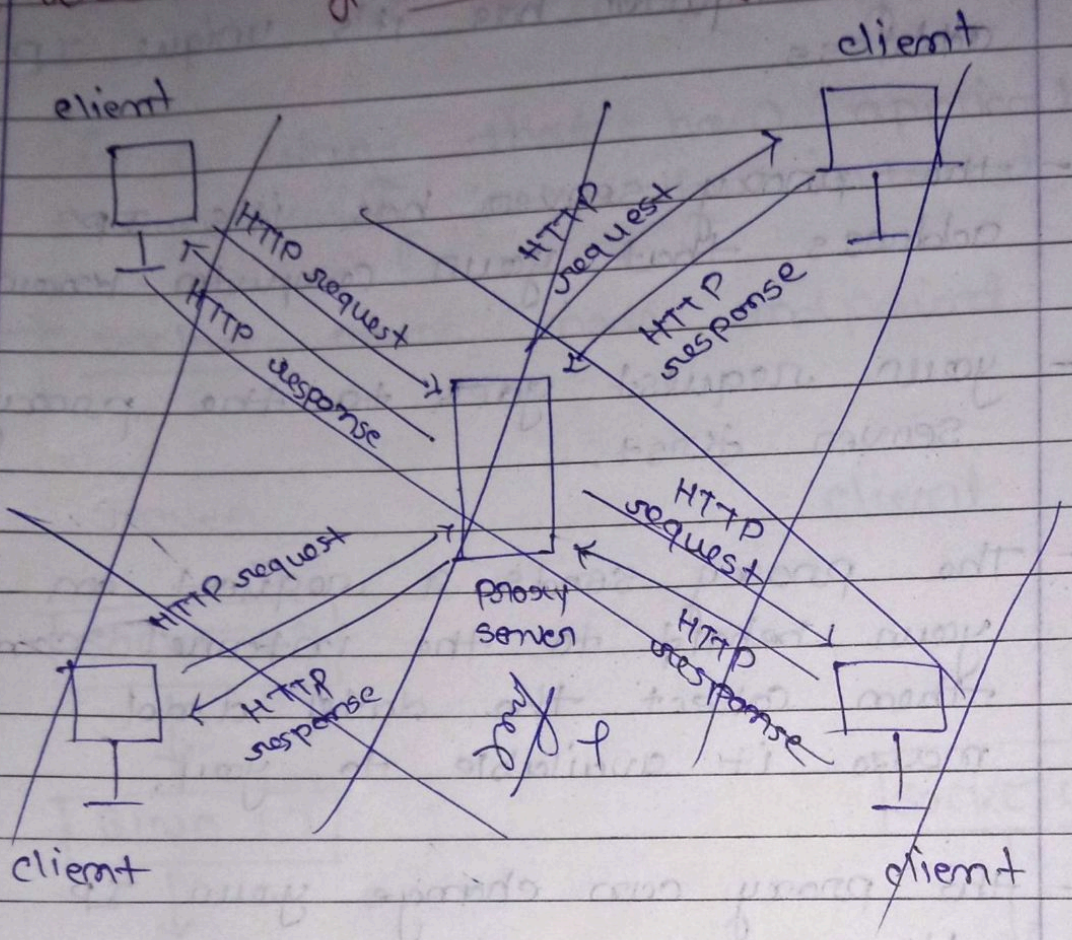
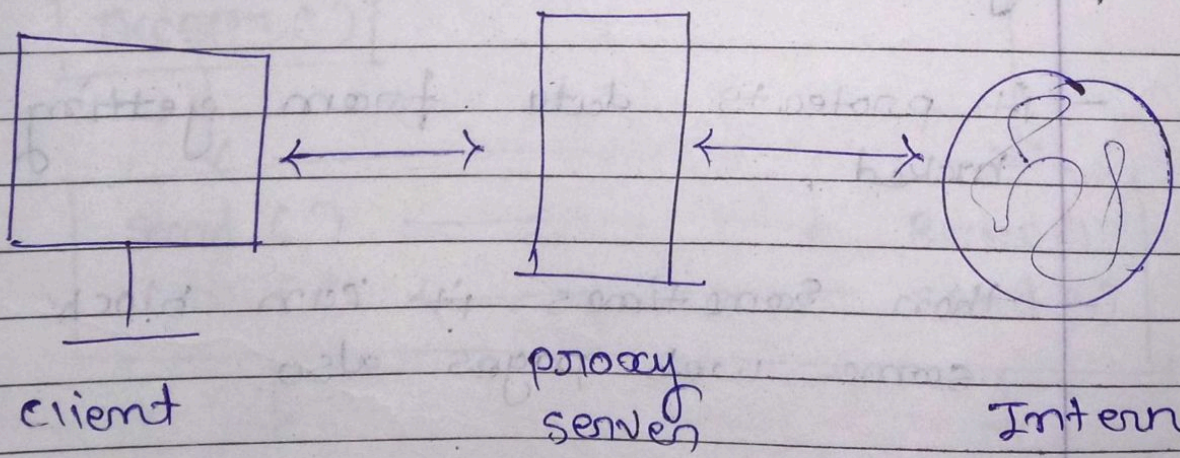
14) Web caching OR proxy server.





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- every computer has it's unique IP address.
- the proxy server has it's IP address that your computer knows.
- your request goes to the proxy server first.
- The proxy sends a request on your behalf to the internet and then collect the data and make it available to you.
- the proxy can change your IP address.
- so, the website will unable to fetch your location in the world.
- it protects data from getting hacked.
- then sometimes it can block some web pages also.

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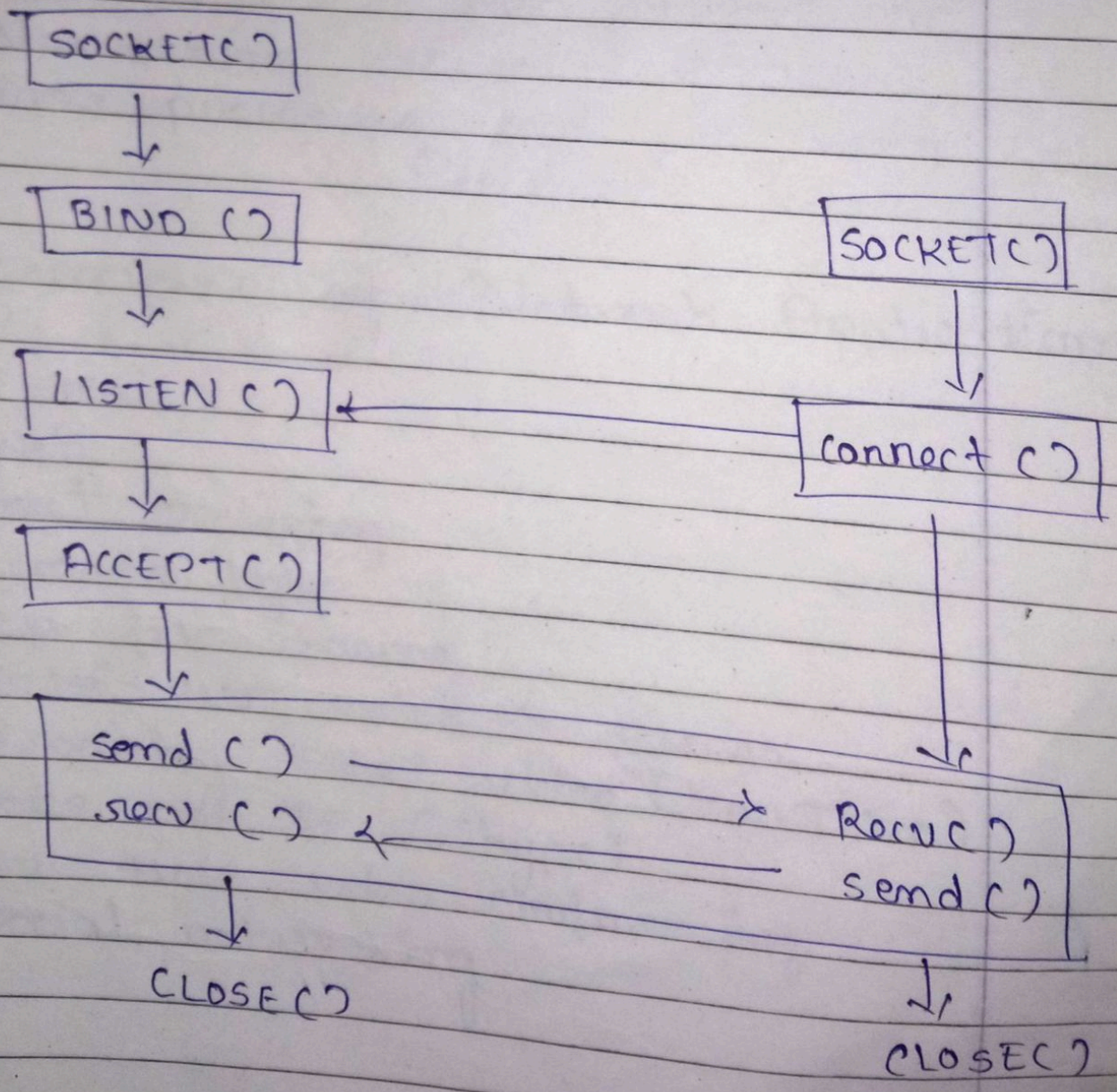
15. Socket programming with TCP & UDP.

is a way that how application communicating over the internet.

→ Socket means node, endpoint.



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Socket (domain , type , protocol)

IPV4	TCP
IPV6	UDP

(TCP) Stream Socket
(UDP) Datagram Socket

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6) principles of computer applications.

=> Computation
- Communication
- Co-ordination
- recollection
- automation
- evaluation and design
- Security
- Input
- output
- Networking
- data processing

* principles of Network Applications.

- e-mail
- Web
- text messaging
- remote login
- p2p file sharing
- multi-user network games
- streaming stored video (YouTube)
- voice over IP (Skype)
- real-time video conferencing
- Social networking.

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Unit :- 3 Transport layer

(1) Transport layer services.

- provides logical communication between application processes running on different hosts.
- implemented in the end systems but not in network routers.
- on sending side, transport layer converts the application-layer messages, known as transport layer segment.
- on receiving side, transport layer reassembles segments into messages.

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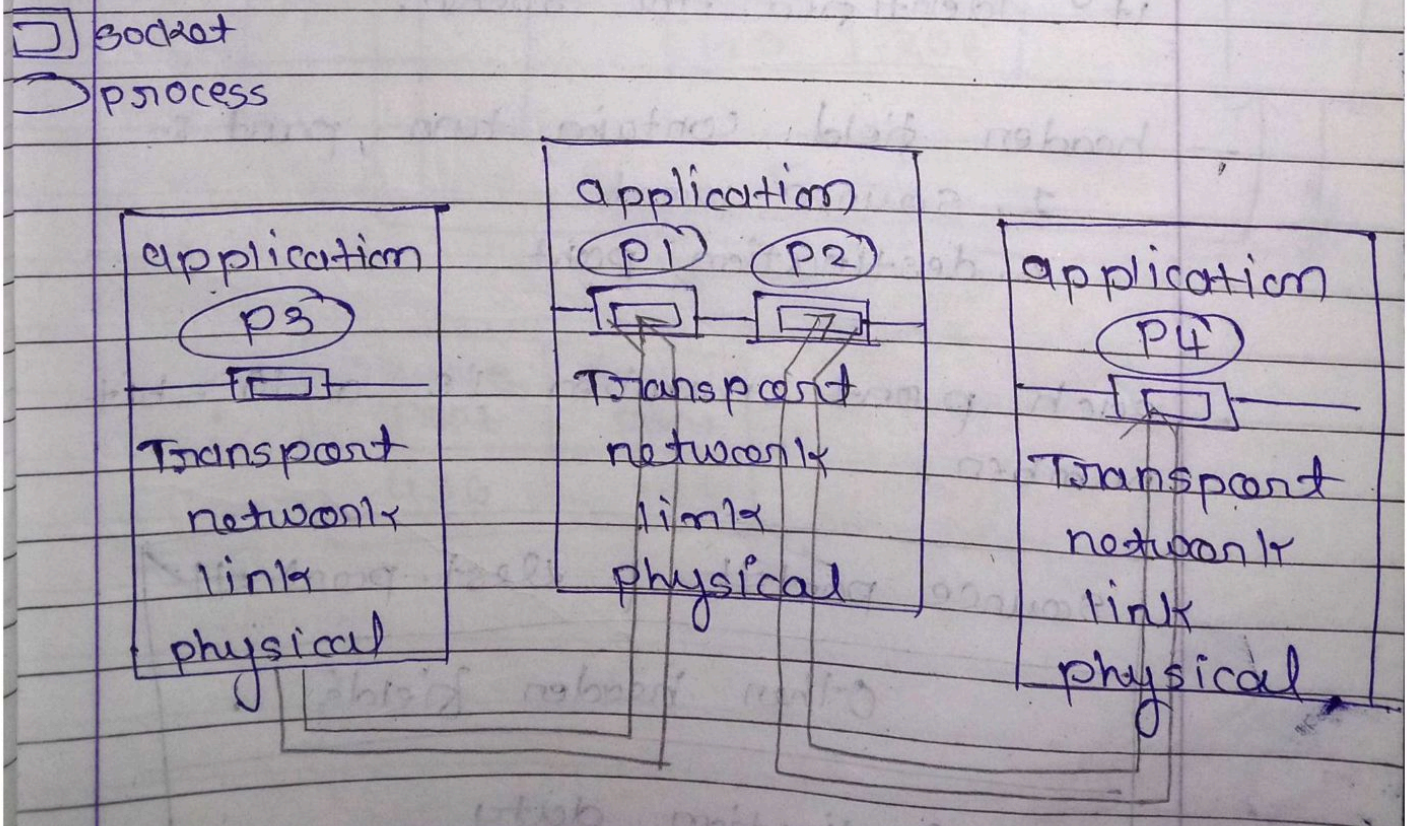
(2) What is Multiplexing and de-multiplexing? Explain it with suitable diagram.

→ Multiplexing :-

Collect data chunks from multiple application process of sender, encapsulating each data chunks with header information and create segment and passing to network layer.

→ Demultiplexing :-

Delivering received segments at receiven side is called demultiplexing.



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- In p1 & p2 process arrive from network layer must be demultiplexed segment.

- gather the outgoing data from socket from transport layer and pass the segment to the network layer.

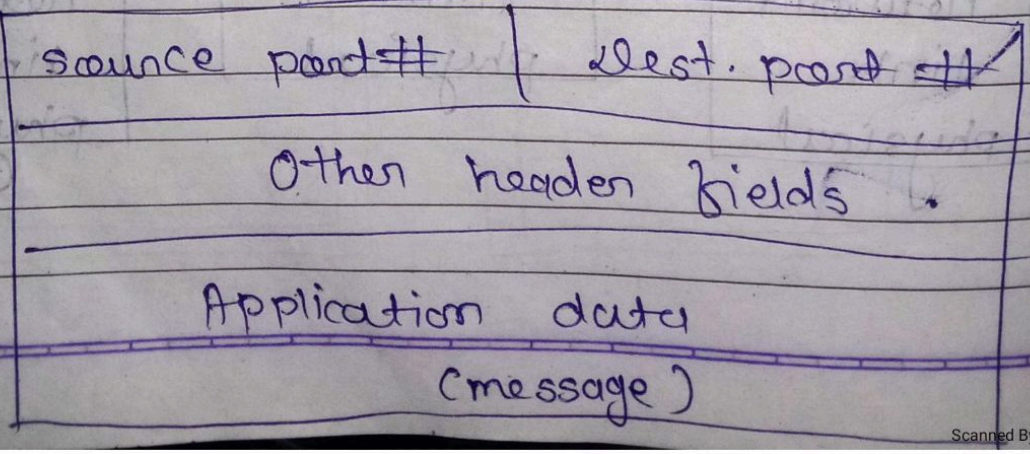
<3> Describe the endpoint identification.

⟹ each sockets must have unique identifire.

— segment include header field and it identified the socket.

- header field contain two part :-
  1. source port
  2. destination port

- each port number is a 16-bit number.



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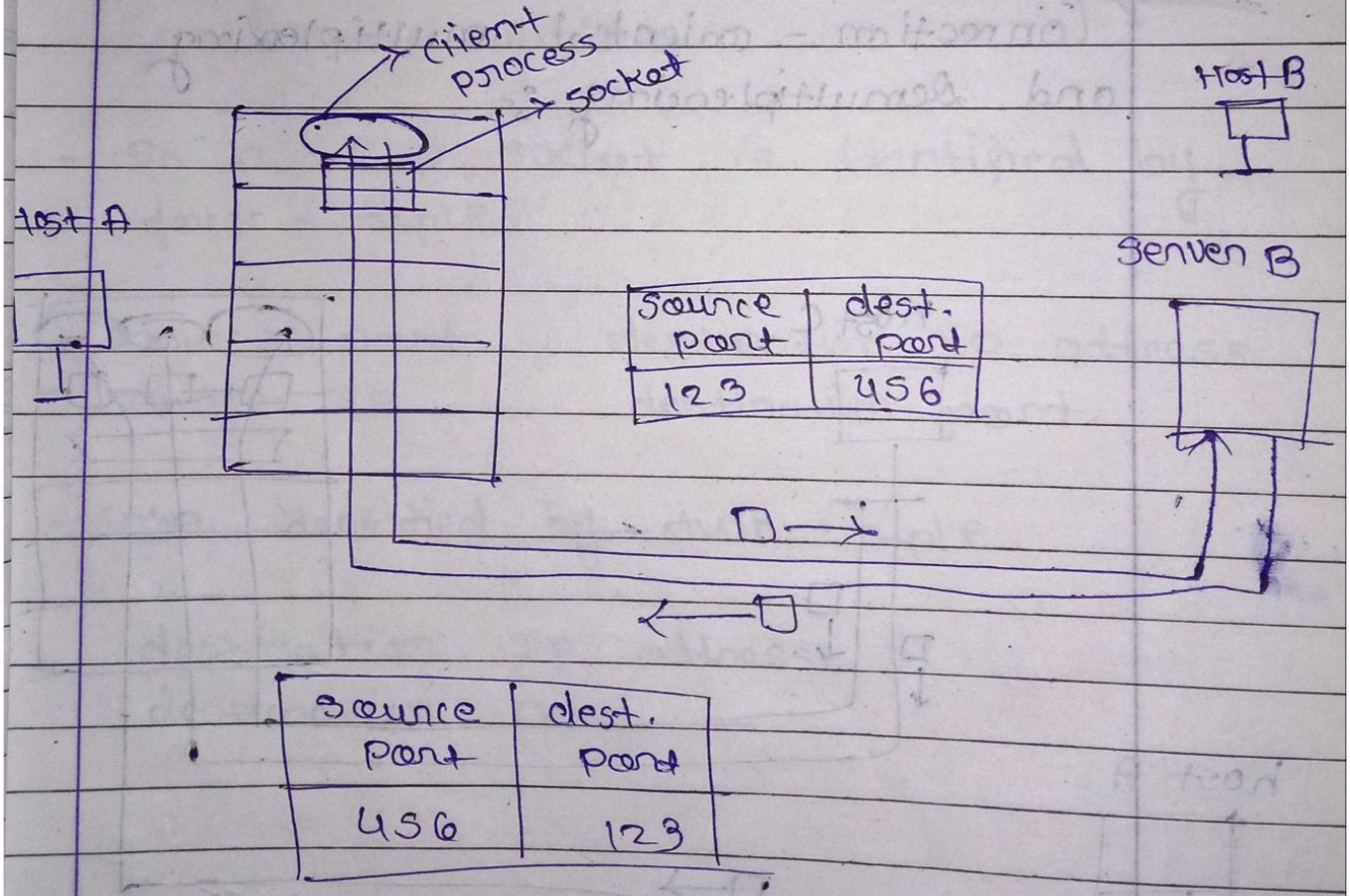
12. Connection-oriented and Connection-less multiplexing and de-multiplexing using suitable diagram.

→ Connection-less multiplexing and de-multiplexing :-

Suppose a process on Host A, with port number 123, wants to send data to a process with UDP port 456 on Host B.



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Header Tcp seg

- each Tcp connection has exactly two end-points.
- Comes two Tcp segments.
- different source Ip address or source port number are two different sockets.
- They have the same destination port number.
- So a Tcp socket is identified by four-tuple.

source port destination IP address
source IP destination port.

- UDP identified by two-tuple.

destination Ip address
destination port.

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Length Tcp seg

- each Tcp connection has exactly two end-points.
- Comes two Tcp segments.
- different source Ip address on source port number are two different sockets.
- They have the same destination port number.
- So a Tcp socket is identified by four-tuple.

source port destination IP address
source IP destination port.

- UDP identified by two-tuple.

destination IP address
destination port.

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(5) Describe the UDP checksum. Explain the each steps of sender and receiver with a 48 bit data segmented as 16 bit segment.

- checksum is a simple error detection mechanism.
- A checksum to verify that the end to end data has not been corrupted by routers or bridges in the network or by the processing in an end system.

Eg.

0 1 0 1 0 1 0 1 0 1 0 1 | 0 1 0 1
1 0 1 0 1 0 1 0 1 0 1 0 | 1 0 1 0
0 1 0 1 0 1 0 1 0 1 0 1 | 0 1 0 1

Sender

0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
+ 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
+ 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 0
1 → 1

0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1

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1's complement

1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0

Receiver

0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1

\+ 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0

1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1

\+ 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1

1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 0

\+ 1

0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1

\+ 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0

1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1

1's complement

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

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{6} Describe the following protocols

(1) Stop and wait :-

- The sender node sends a data packet to the receiver node.
- waits for the feedback of the transmitted packet.
- As soon as the receiver node receives a data packet it starts processing it.
- Then the receiver node sender the feedback to the sender node about the received data packet.
- After receving feedback sender node
- if the feedback is positive there sender node send the next data packet.
- otherwise Resend the damage data packet.

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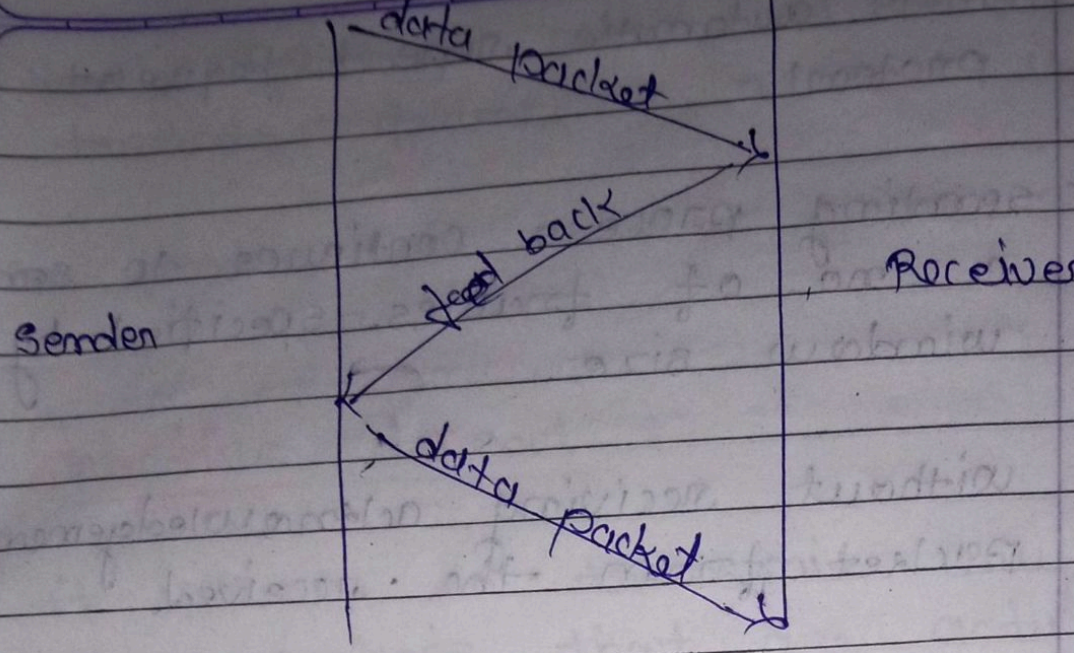
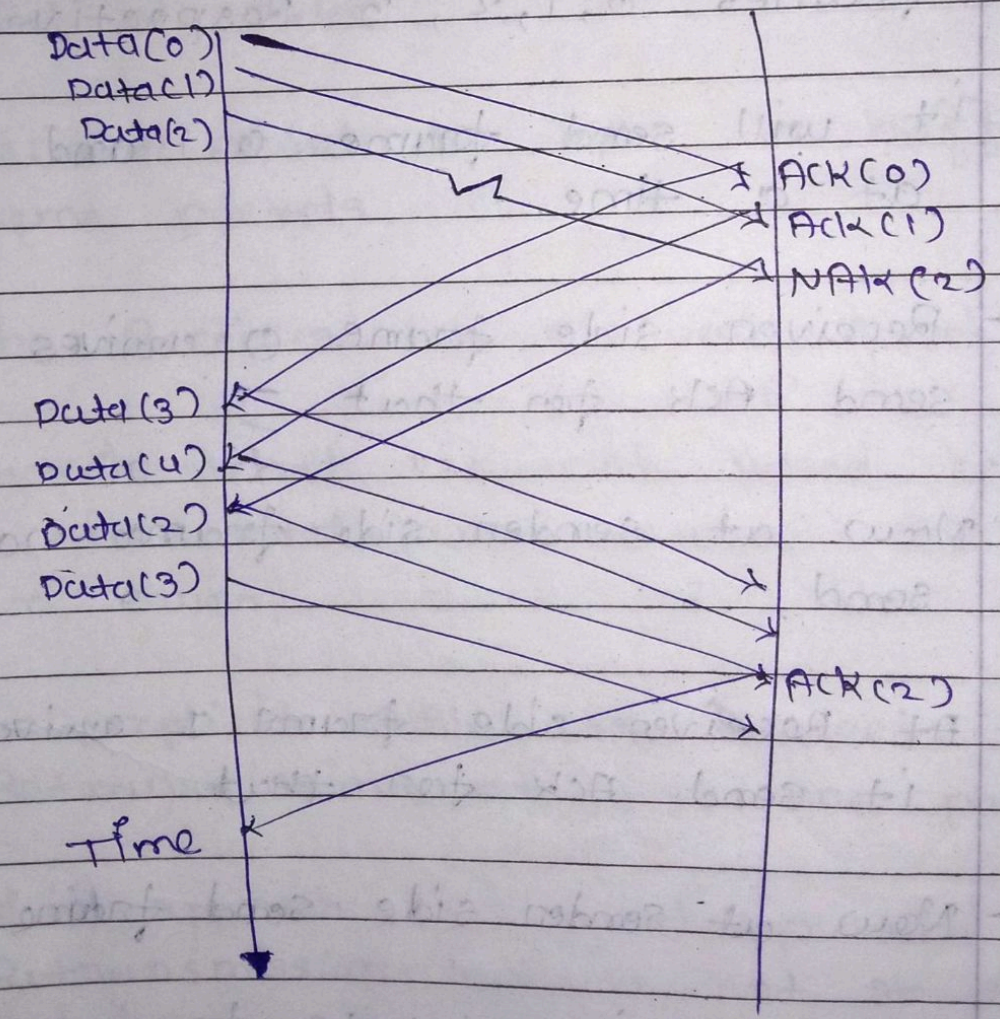


②. Go-back-N :-



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- it is automatic repeat Request protocol.

- sending process continues to send a no. of frames. specified by a window size.

- without reciving acknowledgement packet from the received.

→ working

given example there are four frames 0, 1, 2, 3 Respectively.

- it will send frame 0, 1 and 2 at a time.

- Receiven side frame 0 arrives it send Ack for that

- Now at senden side frame no 3 send.

- At Receiven side frame 1 arrives it send Ack for that

- Now at senden side send frame 4

- Now receiven detects frame 2 is missing or lost.

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- Sender Receiver NAk and sending towards frame 2, 3 and 4.

- So sender will resend frame 2, 3, 4.

3. Selective Repeat :-

- it attemps to transmit only those packets that are actually lost due to errors.

- Receiver must be accept packets out of order.

- Receiver must be able to buffer some packets.

- The Receiver Ack every good packet.

- packets that are not Acked before a time out are assumed lost or in error.

- An explicit NAk can Request Retransmission of just one packet.

- this approach can speed up the Retransmission but is not strictly needed.

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(7) Describe the flow control and Congestion control.

⇒ Flow Control :-

- process of managing the rate of data transmission between two nodes.
- to prevent a fast sender from slow receiver.
- it prevent receiver from becoming overloaded.
- Receiver notice a window with each acknowledgement.
- window
  closed (by sender)
  data is send and Ack'd
  opened (by Receiver)
  data is Read.

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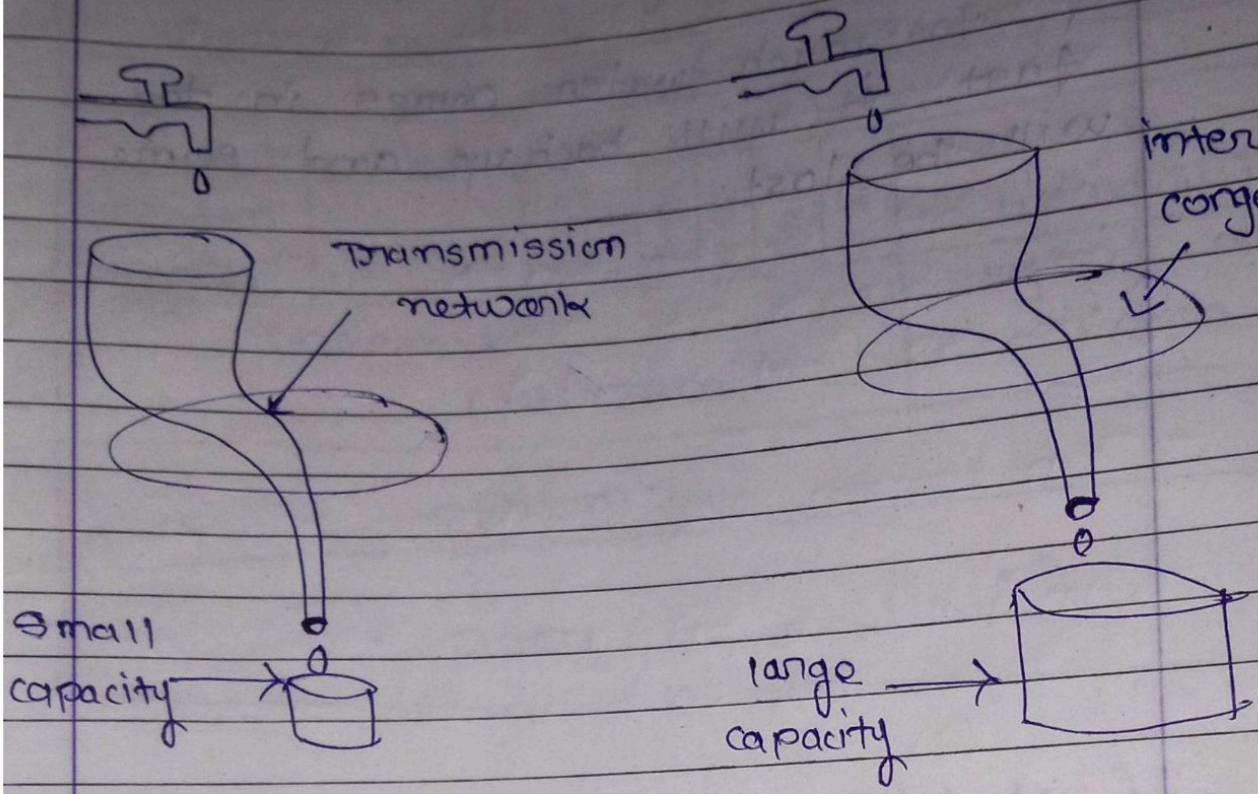
Congestion Control :-



(a) (b).

- in fig. (a), we see a thick pipe leading to a small-capacity receiver.
- As long as the sender does not send more water than the bucket can contain, no water will be lost.
- in fig (b), the limiting factor is not the bucket capacity, but the internal carrying capacity of the network.
- fig. (a), a fast network feeding a low capacity receiver.

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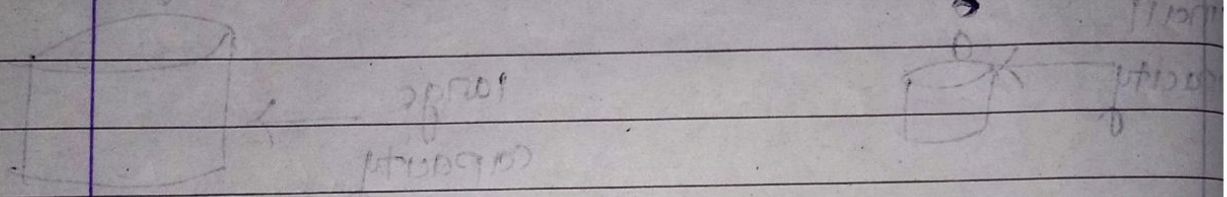
- fig. (b), A slow network feeding a high-capacity receiver.
- if too much water comes in too fast, it will back up and some will be lost.



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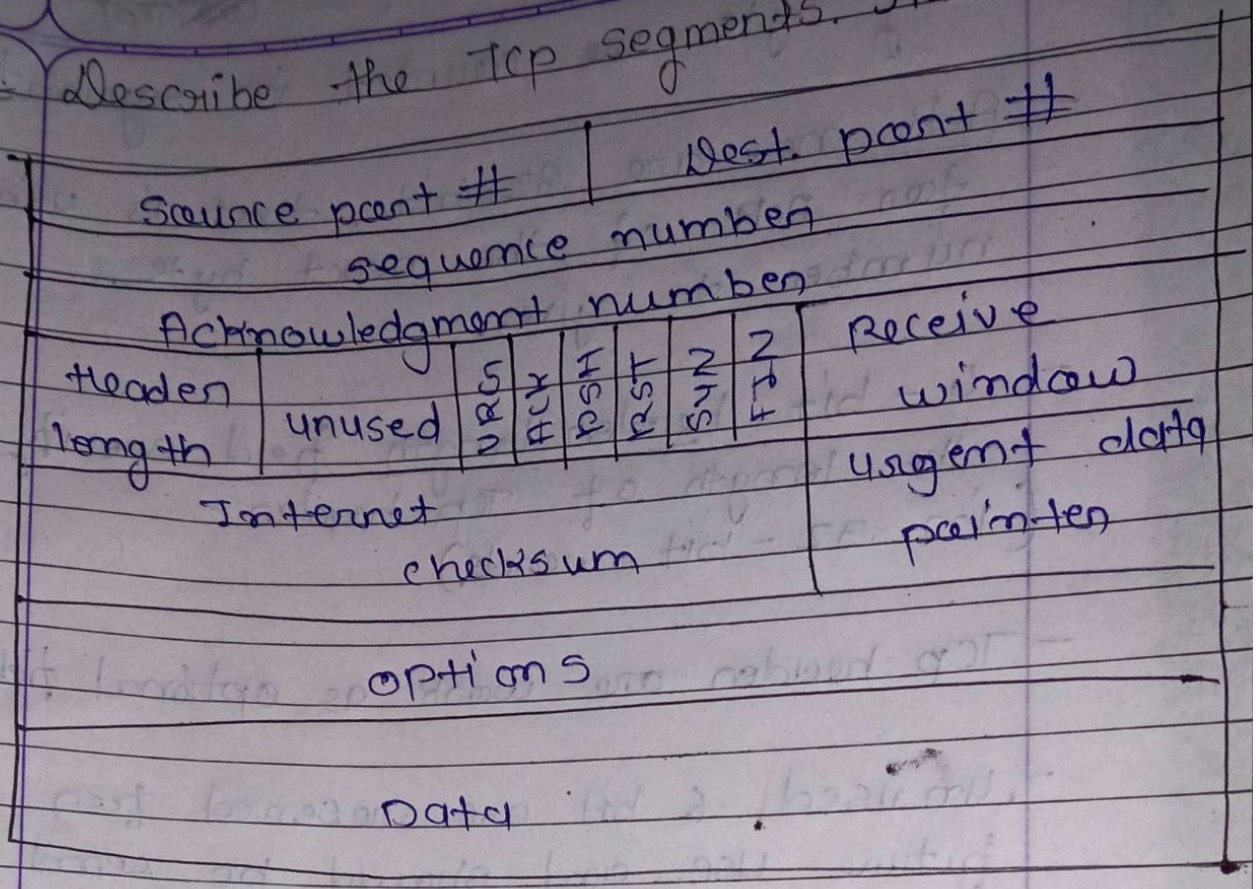
1. Describe the Tcp segments structure.



- Tcp header have (min - 20 bytes) (max - 60 Bytes)
- data are transmission in Tcp is called segment.
- Header include source & destination port number. (both are 16 bit)
- it is use for multiplexing - demultiplexing Data from to upper layer application
- 32 - bit sequence number field that sequence no. of a segment is the byte string no. of the first bit in the segment.

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(16-bit) • checksum : error checking.

(16-bit) • window field used for flow control

- every byte are counted in segment.
- Ack no. are 32-bit, that is use for Ack of the next byte number to give.
- 4-bit header length field specify the length of TCP header. (in 32-bit words.)
- TCP header are work as optional field.
- Unused 6 bit are reserved for future use and should be send to 0.

(1 bit) • URG : we can transmit the urgent data and set 1.

• Ack : that indicate the Ack field. (1 bit)

(1bit) • PSH : push function, asked to push the buffer data to the receiver application.

(1 bit) • RST : Reset the connection.

(1 bit) • SYN : synchronize the sequence of a number.

(1 bit) • FIN : No more data from sender.

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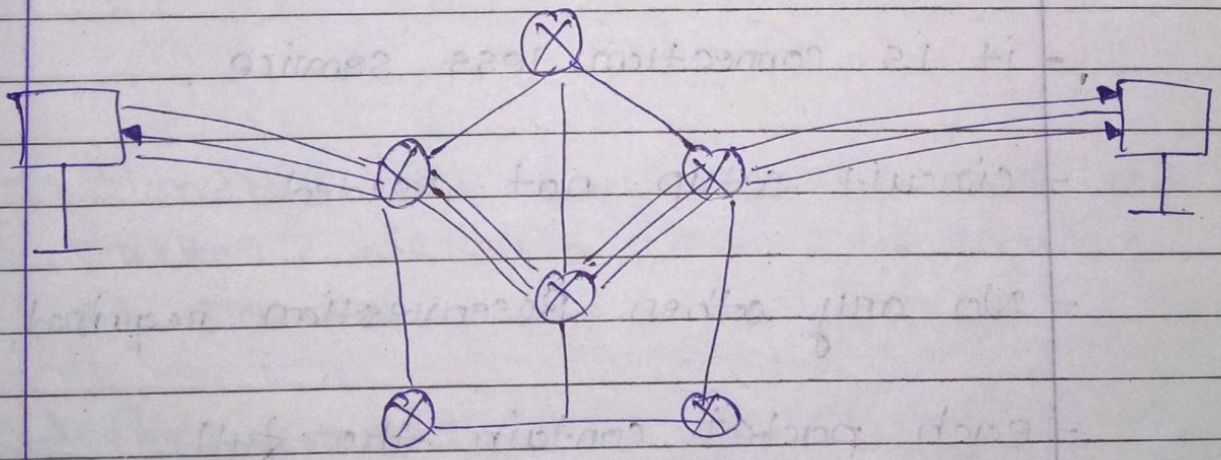
(1) Describe the following :

(a). Virtual Switching :-

- It is connection oriented service.
- Entire message have follow same router during transmission.
- Circuit setup Required.
- State information about each VC is maintained.
- Resource reserve for the time interval of data transmission between two nodes.



→ Working :-

- medium is set between two end nodes.

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- Reserve the Resource for transmission packets.

- signals are send to sender and inform the all medium is set up to transmission can be started.

- ensures the transmission of all packets.

- global header is used in the first packet of the connection.

- whenever data is to be transmitted a new connection is setup.

(b). Datagram network :-

- it is connection less semice.

- circuit setup not needed.

- No any other Resenvestion required.

- each packet contain the full source and destination address.

- Routers do not hold state information about connections.

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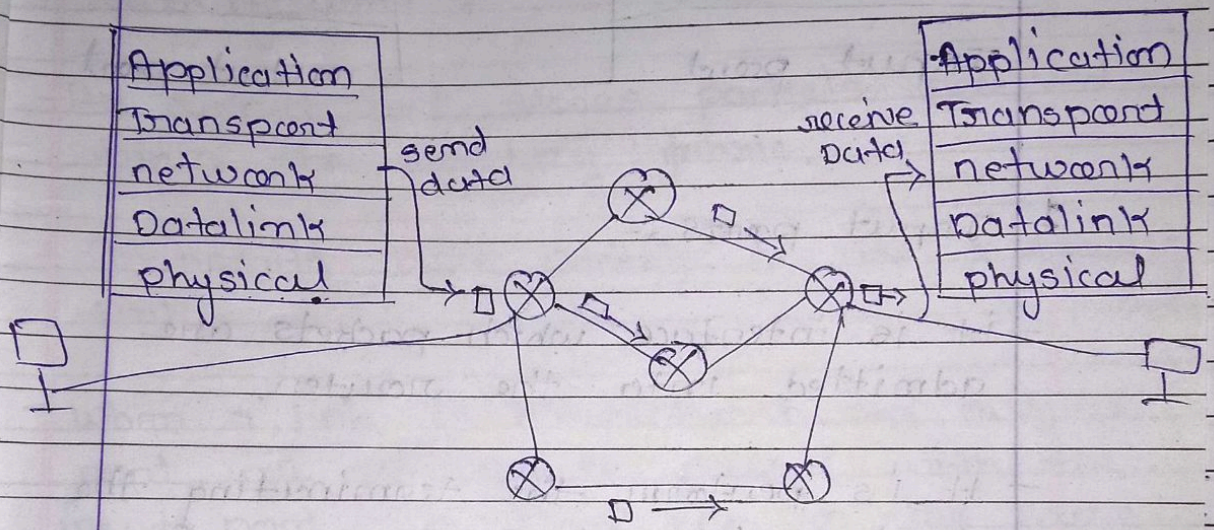
- packet Routed independently.
- all packets routed independently router resource requirement can vary.
- if one router are crash, them only those side of packet are fail.

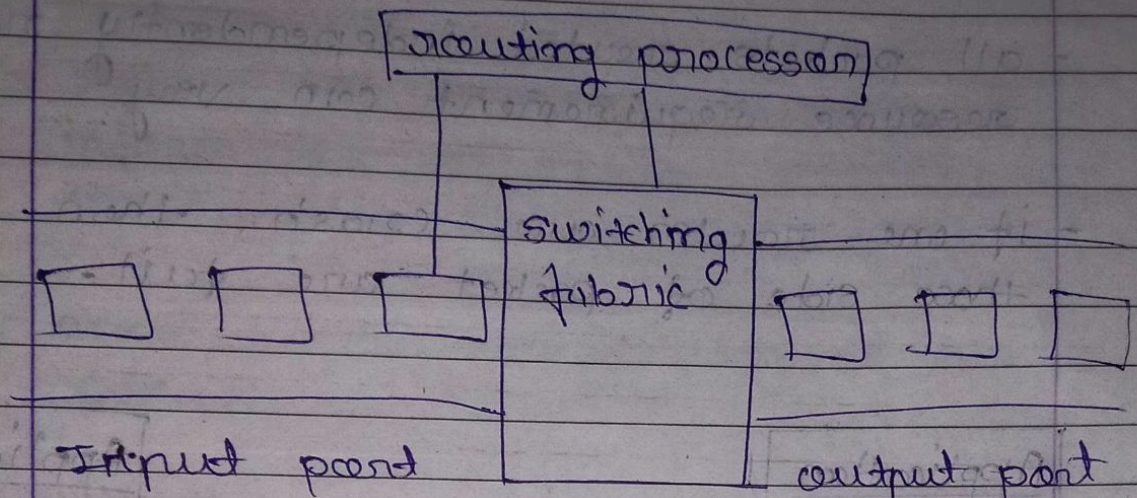


Q2. Mention the four components of router. Describe the architecture of router.

→ Components :-

1. Input ports
2. switching fabric
3. output ports
4. Routing processor

Architecture :-



1. Input ports :-

- it is interface which packets are admitted into the router.
- it is perform the terminating the physical link at Router.
- it's done by left part of figure. The middle part does the work of interoperating with link layer like decapsulation.
- last part of ilp port the forwarding table is looked up and is used to determine the appropiate olp port based on destination address.

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2. Switching fabric :-

- The switching fabric connects the router's i/p ports to its o/p ports.
- This switching fabric is completely contained within the router.

3. output ports :-

- An output port stores packets received from the switching fabric.
- Transmits these packets on the outgoing link.
- When a link is bidirectional, an output port will typically be paired with the input port for that link on the same line card.

4. Routing processor :-

- it is executing Routing protocol.
- maintain Routing tables.
- attach link state information and computes the forwarding table for the router.
- nlw management function performe.

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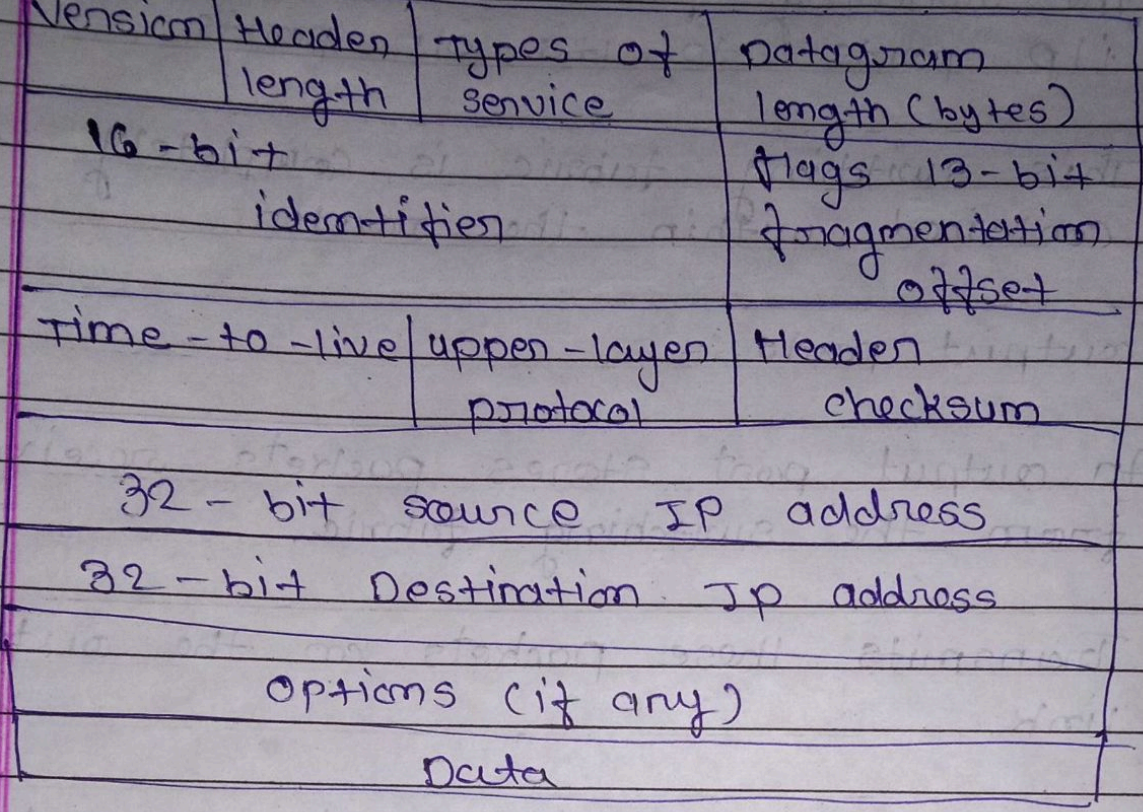
{3} IPV4 datagram.



- it is connectionless protocol used for packet-switched networks.
- it is use 32-bit addresses.

- Version number :-

- it is 4 bit that specify the IP protocol version of the datagram.
- only permitted value 4 (0100) or 6 (0110).

- Header length :-

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- it is specify the length of Ip header in 32 bit words.

- minimum 5 , maximum 15

- Types of service :-

- that allow different type of service low delay, high throughput, Relibility.

- 8-bits

- Identification :-

- unique packet id for identifying the group of fragments of a single Ip datagram.

- Total length :-

- length of header + data (16 bits)

- minimum value 20 bytes and maximum value 65,535 bytes.

- flags :-

- 3 flag for 1 bit each.

- Reserved bit, do not fragment flag, more fragment flag.

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- fragmentation offset :-
  - fragmentation of a message occurs.
  - this field specifies the offset or position.
  - overall message when data is fragment goes.
  - specified units of 8 bytes.
- time - to - live :-
  - specifies how long the datagram is allowed to "live" on the network.
- protocol :-
  - name of the protocol to which the data is to be passed. (8 bits)
  - which is use TCP, UDP.
- Headen checksum :-
  - 16 bits headen checksum for checking error in the datagram header.
- Source Ip address :-
  - 32 bits IP address of the sender.

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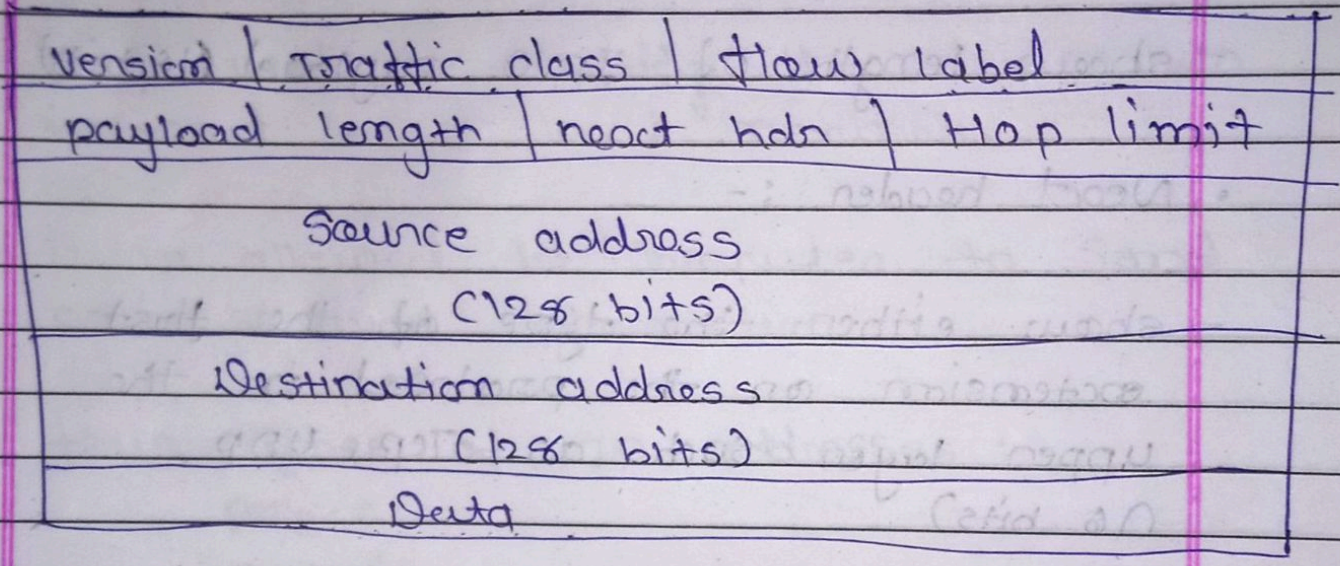
- destination Ip address :-

- 32 bits Ip address of the receiver.

- option :-

- allows Ip header to be extended.

(4) IpV6 datagram.



- Version :-

- Version field shows the version of IP and set to 6.

- Traffic class :-

- it is 8 bits.

- it is similar to the IpV4 type of service filed.

- indicate IpV6 packet's class or priority.

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- flow label :-

- provide additional support for Real-time datagram delivery and quality of service feature.

- specify sequence of flow packets between source to destination. (20 bits)

- payload length :-

- show length of IPV6 payload (16 bits).

- Next header :-

- show either the type of the first extension or the protocol in the upper layer such as TCP, UDP. (16 bits)

- hop limit :-

- shows the maximum no. of Routers the IPV6 packet can travel. (8 bits)

- source address :-

- source of packet address (128 bits)

- destination address :-

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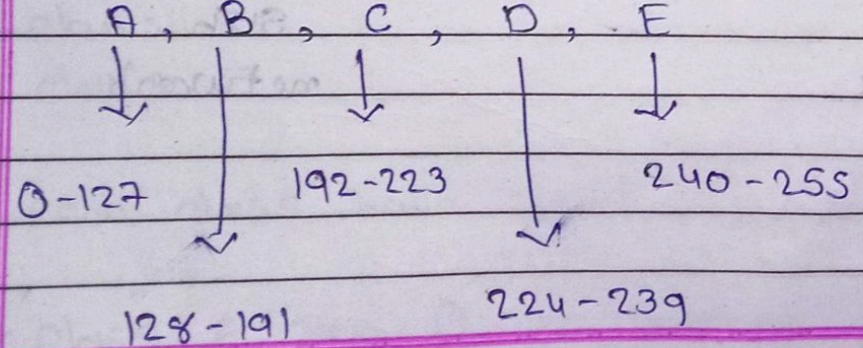
- show the address of the destination of the packet (128 bits).

5). Describe the following terms.

(1). Ip :-

- Ip address is internet protocol.
- it is unique address.
- it is used to identify the device on internet on a local network.
- it is allowed to computer to send and received information.
- four types of Ip addresses :
  - public
  - private
  - static
  - dynamic
- five class of Ip addresses:



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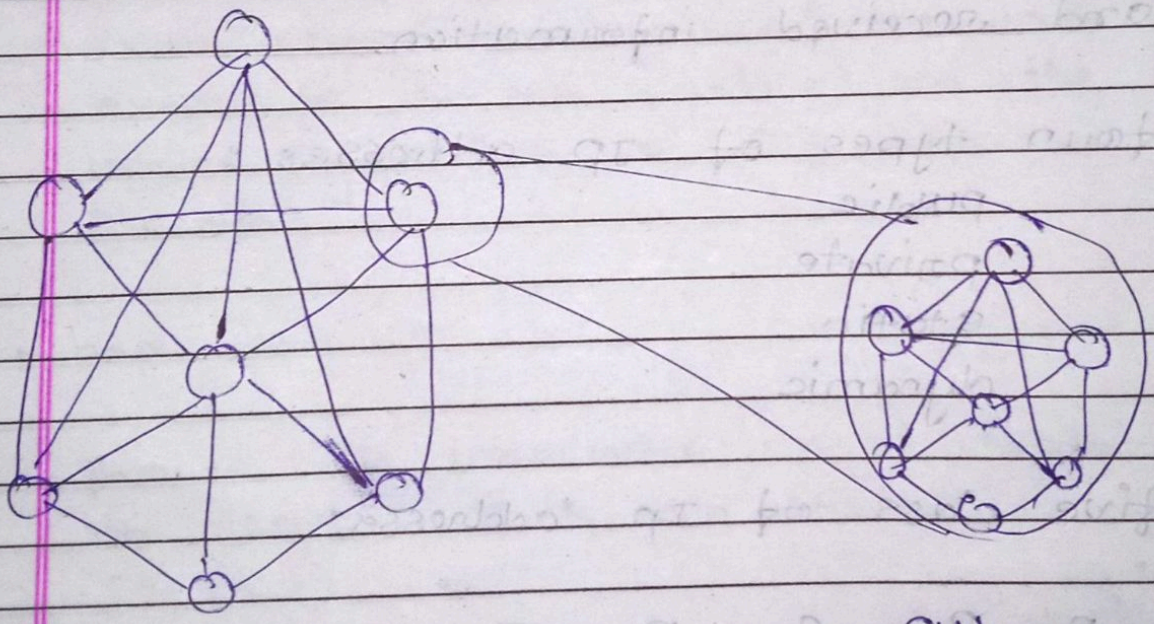
(2). Subnet :-

- When the bigger network devided into smaller networks.
- Order to maintain security, maintanance is easier for small networks.
- it is more efficience, performace high.
- we can travel short distance without passing an necessary Routers to Reach it's destination.



network

sub network

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(6) Describe all the classes of classful addressing.

- Ip address is 32-bit unique address
- address space of 2^{32}
- There are two notations in Ip address is written, dotted decimal, hexadecimal notation.

→ Dotted

128 . 11 . 3 . 31

10000000 00001011 00000111 00011111

- The value of any segment is between 0 to 255.
- 32-bit address dividing into five sub classes.

Class A
Class B
class C
class D
class E

- each class have valid Ip address range.
- Class E and D are Reserved for multicast experimental Purpose.

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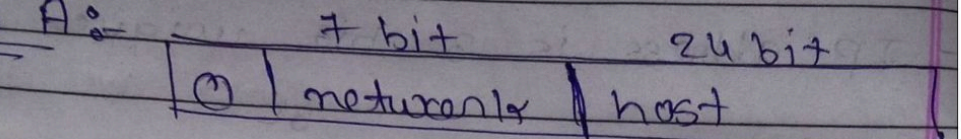
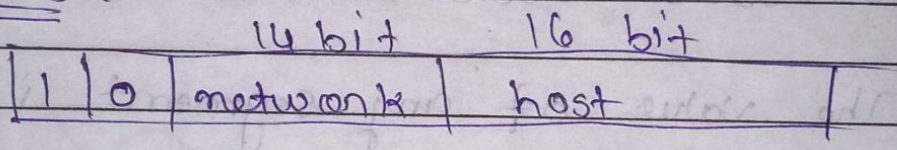
- network Id, host Id contain in classes.

⟹ class A :-



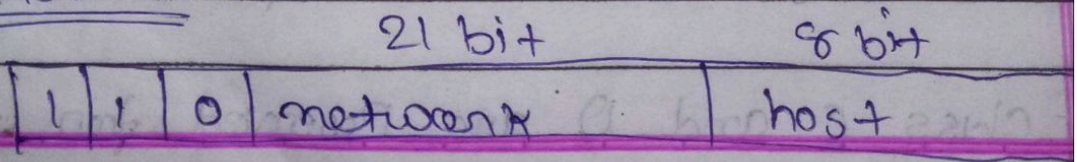
- large no of hosts assigned
- higher bit first octel always set to
- 7 bits for used determine network id
- 24 bits for used determine host in any network.

⟹ class B :-



- higher bit set to octel 10,
- 14 bits are used to determine network id.
- 16 bits of host id in any network.
- medium size to large size network assigned.

⟹ class C :-



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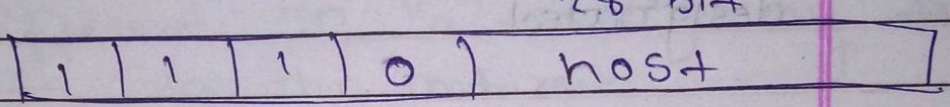
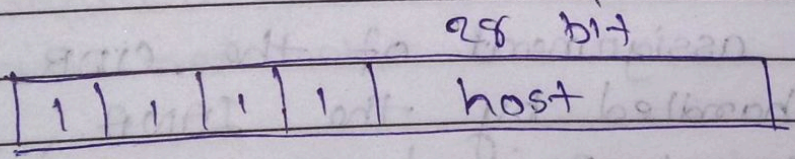
- assigned to small sized network.
- higher order first octet set to 110.
- 21 bit used to network id.
- 8 bit of host id to determine host in any network.

⟹ class D :-



- Reserved for multi-casting.
- higher order bits set to 1110.
- does not posses any subnet mask.

⟹ class E :-



- E are Reserved for experimental and Research purpose.
- doesn't have any subnet mask.

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(7) Describe the CIDR.

(Classless Inter-Domain Routing.)

- It is an Ip address assigning method that improves the efficiency of address distribution.

- By using a single CIDR Ip address many unique Ip addresses can be designated.

- CIDR Ip address except that it ends with a slash.

172.200.0.0/16
It called Ip network prefix.

- CIDR provide efficient address space allocation.

- The assignment of the CIDR block is handled by the IANA.
(Internet Assigned number Authority.)

- No separate subnetting method is required.

- It allows efficient routing entries.

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{8} Describe the following.

DHCP

- (Dynamic Host Configuration Protocol.)
- used to assign dynamic Ip addresses to devices on a network.
- with dynamic addressing, a device can have a different Ip address every time it connects to the network.
- In some systems, the device's Ip address can even change while it is still connected.
- implemented on local networks as well as large network.
- also support mobile users who want to join network.
- There are many versions of DHCP are available for use in IPV4 and IPV6.
- Dhcp maintains the unique Ip address of the host using a DHCP server.

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NAT

(Network Address Translation.)

- The idea of NAT is to allow multiple devices to access the Internet through a single public address.

- To achieve this, the translation of a private Ip address to a public Ip address is required.

- NAT is process in which one or more local Ip address is translated into one or more Global IP address.

- Also, it does the translation of port numbers.

- NAT generally operates on a router or firewall.

- There are 3 ways to configure NAT:
  1. Static NAT
  2. Dynamic NAT
  3. port address Translation

- it provides privacy.

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ICMP

(Internet Control message protocol.)

- it is a network layer protocol.
- used for error handling.
- and it is primarily used on network devices such as routers.
- ICMP can be used to report these errors and debug those errors.
- In this case, the router sends the message to the sender that I could not send the message to that destination.

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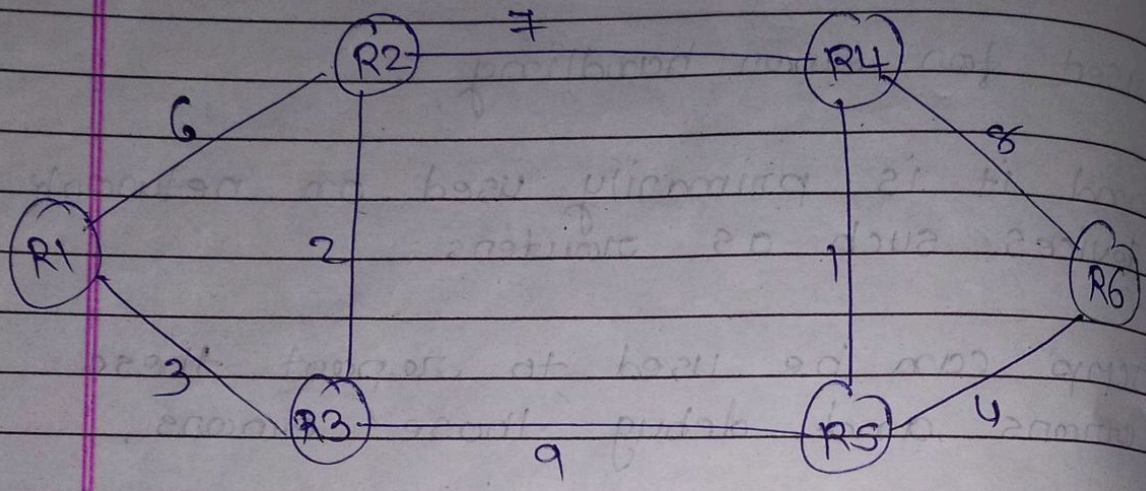
(9) Link - State Routing (Dijkstra's algorithm)



link state table

R1	
R2	6
R3	3

R3	
R2	2
R5	9
R1	3

R5	
R3	9
R4	1
R6	4

R6	
R5	8
R4	4

R2	
R1	6
R3	2
R4	7

R4	
R2	7
R5	1
R6	8

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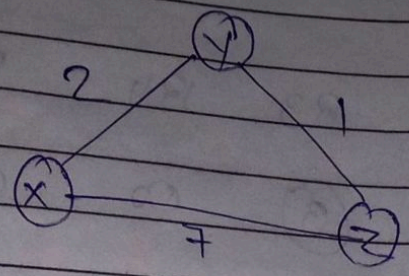
Routing table

R1	R2	R3	R4	R5	R6
	6	(3)	∞	∞	∞
R1 R3	(5)	(3)	∞	12	∞
R1 R3 R2			(12)	12	∞
R1 R3 R2 R4				(12)	21
R1 R3 R2 R4 R5					16

→ computes the least-cost path from one node to all other nodes in the network.

→ In this algorithm, each router in the network understands the new topology then makes a routing table.

Distance Vector routing



	x	y	z
x	0	2	7
y	∞	∞	∞
z	∞	∞	∞

	x	y	z
x	0	2	3
y	2	0	1
z	7	1	0

	x	y	z
x	0	2	3
y	2	0	1
z	3	1	0

	x	y	z
x	∞	∞	∞
y	2	0	1
z	∞	∞	∞

	x	y	z
x	0	2	7
y	2	0	1
z	7	1	0

	x	y	z
x	0	2	3
y	2	0	1
z	3	1	0

	x	y	z
x	∞	∞	∞
y	∞	∞	∞
z	7	1	0

	x	y	z
x	0	2	7
y	2	0	1
z	3	1	0

	x	y	z
x	0	2	3
y	2	0	1
z	3	1	0

- it is iterative, distributed & asynchronous
- dynamic algorithm.
- each router maintain distance table.
- broadcast update.

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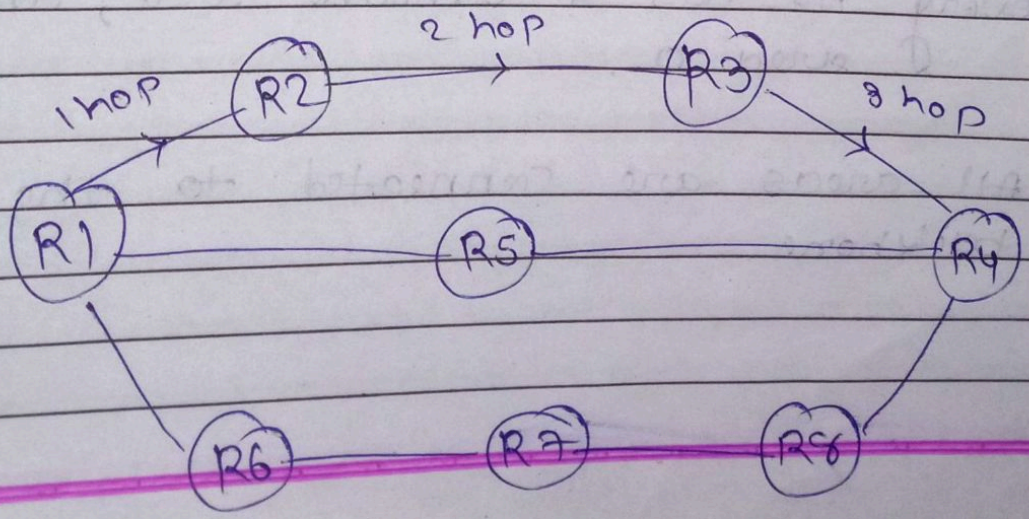
Q10. Describe the protocols.

RIP

(Routing Information protocol.)

- Connect networks using the IP, to share information about how to route traffic among nlw.
- Each RIP router maintains a routing table.
- RIP uses a distance vector algorithm to decide which path to put packet on to get to its destination.
- Using RIP, each router sends its entire routing table to its closest neighbour every 30 seconds.
- When the router sends the packet to the network segment, then it is counted as a single hop.



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- RIP can support maximum upto 15 hops.

- which means that the 16 routers can be configured in a RIP.

OSPF

(open shortest path first.)

- has been designed within a single autonomous system. (AS)

- it is an intradomain proto(o), which means that is used within an area or a network.

- OSPF achieves by learning about every router and subnet within the entire network.

- OSPF uses link state

- every AS has a backbone area, called area 0.

- All areas are connected to the backbone.

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Unit :- 5 Datalink layer

Q1. Describe the ethernet and it's various type.

⟹ Ethernet :-

- one of the widely used local area network technology.
- it connects computers within the local area network and wide are network.
- it offers a simple user interface that helps to connect various devices easily.

⟹ Types of Ethernet :-

(1). switched Ethernet
(2). fast Ethernet
(3). Gigabyte Ethernet

(1). switched Ethernet :-

- it is indicated 10 mbps bandwidth on each of its ports.
- each port one can connect to a thin segment or a computer.
- it is collision domain is separated
- hub is Replaced by a switch.

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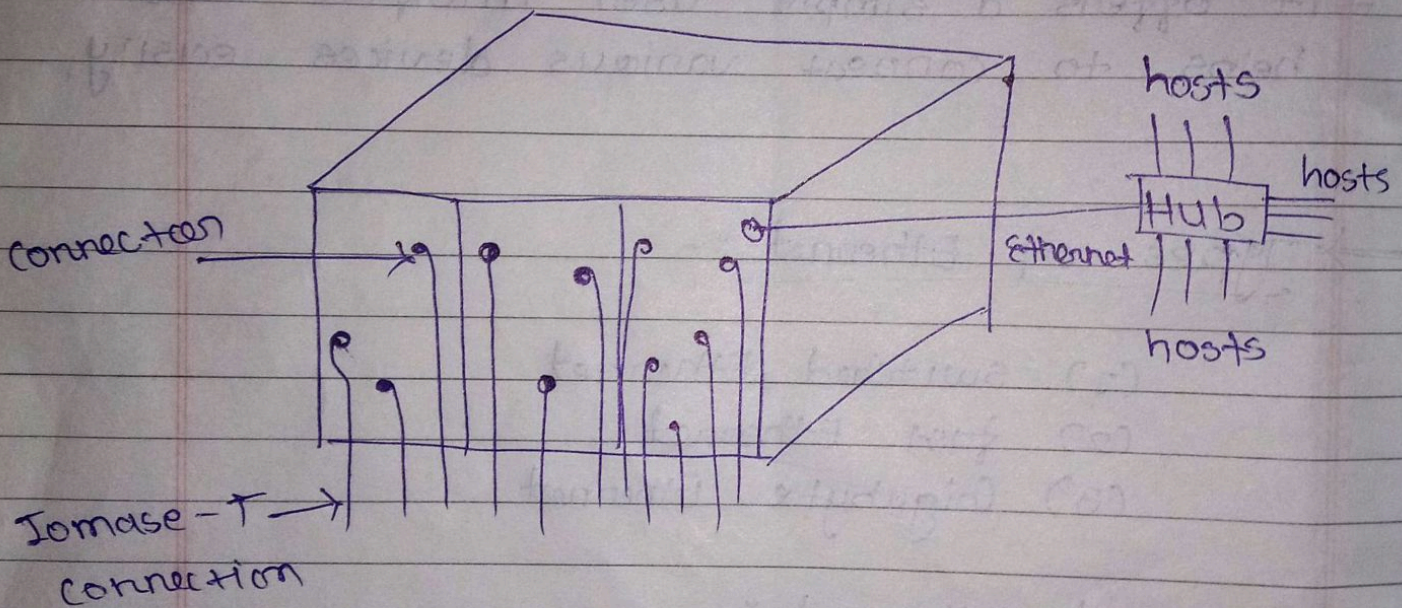
- destination address of the Received frame and forward the frame to the port to destination station is connected.

- it is physical and logical topologies are star.

- increased on switched ethernet by using full-duplex technique.



(2). fast Ethernet :-

- it is approved by IEEE Committe.
- it is high speed internet.
- it can transfer and Receive data 100 mbps.

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- usually supported by a twisted pair or CAT5 cable.

- we connected laptop, camera any other device is connected to network.

→ two type categories : 1. 100Base-X
2. 100Base-T4

- it keep the same frame format.

- keep the 48-bit address.

(3). Gigabyte Ethernet :-

- it is based on fiber optical cable.

- multi-mode fiber is able to transmit rate to at least 550 meters.

- it is costly.

- it is higher speed data transmit.

- 1000 mbps or 1Gbps.

- in twisted pair all pairs cables constribute to the data transfer speed.

- use to large application in video calling system.

- use multimode and single mode system.

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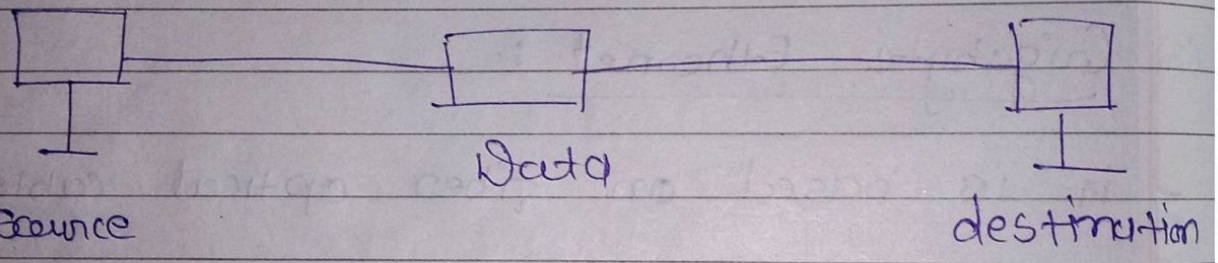
12. ARP protocol.

- (Address Resolution protocol.)
- mapping Ip address to physical address.
- Example is Ipv4 version, it most common use today.
- 32 bit long.

→ How ARP work :



1. host 1 broadcast ARP request to all devices.
2. Compose the query.
3. all device are compare the Ip address to other device.
4. if ip address match then send ARP responce.
5. then transmit the packet to destinati

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msb → add front
lsb → add last

(3) What are the various error detection techniques ?, and correction.

⟹ Techniques for error detection
1. parity checks
2. checksum methods
3. cyclic redancy checks

(1). parity checks :-

- single parity checking is the simple mechanism and inexpensive to detect the errors.

- 1 is added if it contains odd number of 1's, and

0 is added if it contains even number of 1's.

Eg

odd

original data

0	1	0	1	0	0	1

no. of 1 is odd
total no. of odd is = 3
we can add last to 1.

0	1	0	1	0	0	1	1

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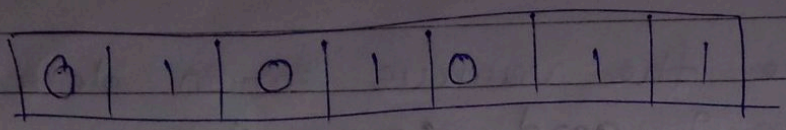
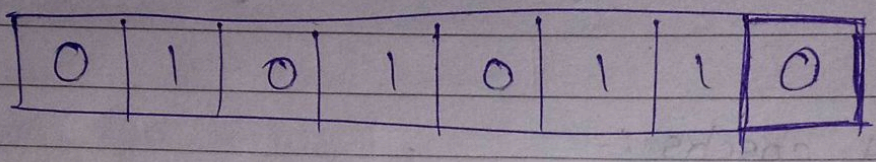
evem



no. of 1 is evem
total even no. is = 4
we can add last to 0



(3). CRC :-

- (cyclic Redundancy checks).
- It is most powerful and easy to implement technique.
- CRC is based on binary division.
- Sequence of Redundant bits, are appended to the end of data.
- The Resulting data unit becomes exactly divisible by a second, predetermind binary number.
- If at this step no Reminder, the data is assumed to be correct and is therefore accepted.

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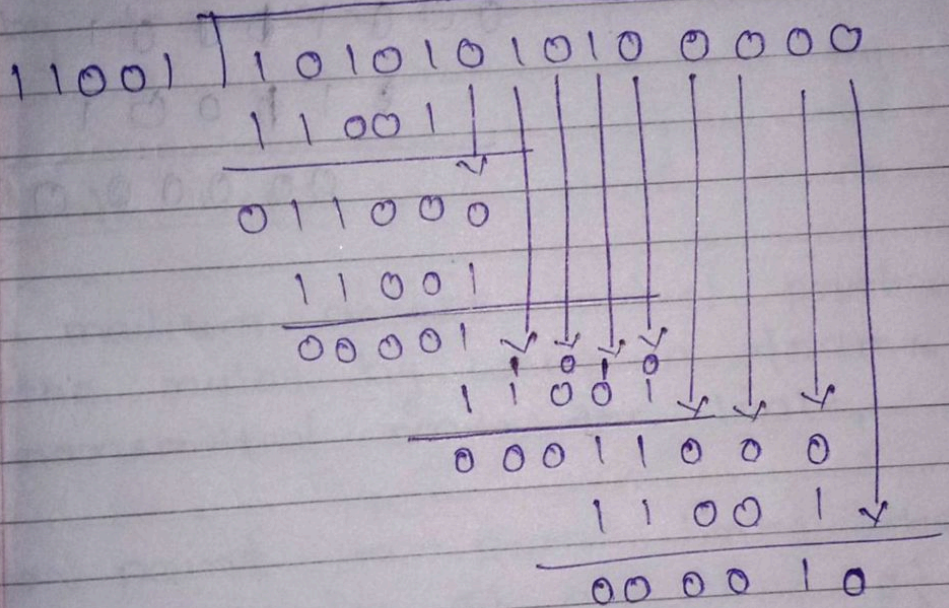
- A Reminder indicates the data is demaged and Reject to data.

Eg: 1010101010

$x^4 + x^3 + 1$

1 1001

sender



10101010100010

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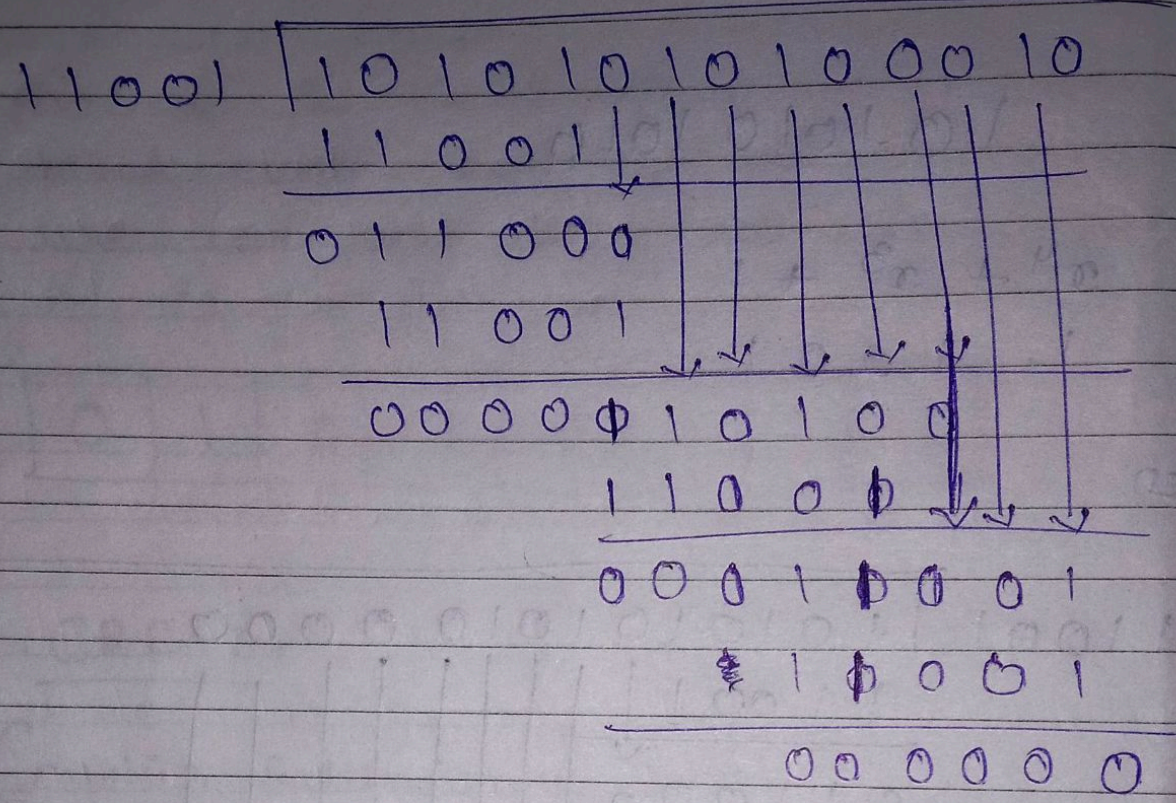
Receiver



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{4} Describe the service provided by link layer.

- Framing :-

- Almost all link-layer protocols encapsulate each network-layer datagram within a link-layer frame before transmission over the link.

- The structure of the frame is specified by the link-layer protocol.

- Link access :-

- A medium access control protocol specifies the rules by which a frame is transmitted onto the link.

- for point-to-point links that have a single sender at one end of the link.

- the MAC protocol is simple the sender can send a frame whenever the link is idle.

- Reliable delivery :-

- When a link-layer protocol provides reliable delivery service,

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- it guarantees to move each network layer datagram across the link without error.

- A link-layer reliable delivery service can be achieved with acknowledgments and retransmission.

- Error detection and correction :-

- The link-layer hardware in a receiving node can incorrectly decide that a bit in a frame is zero when it was transmitted as a one, and vice versa.

- such bit errors are introduced by ~~sing~~ signal attenuation and electro-magnetic noise.

- Because there is no need to forward a datagram that has an error.

- many link-layer protocols provide a mechanism to detect such bit errors.

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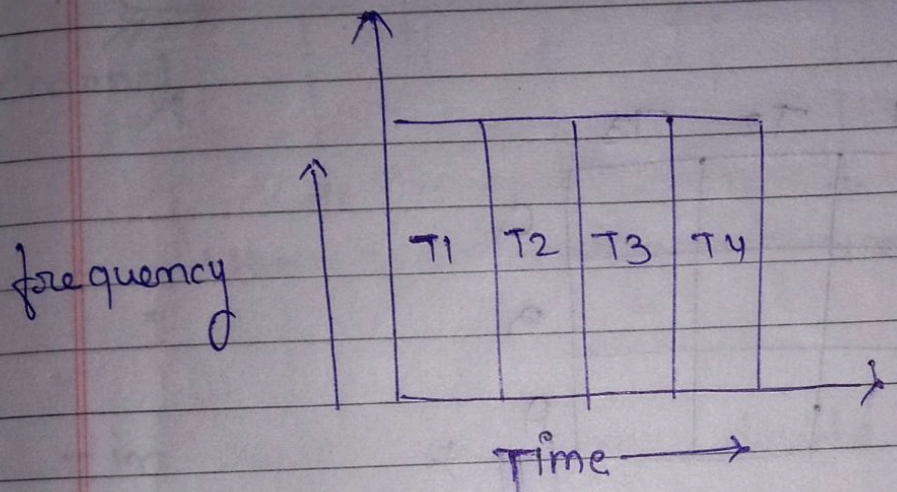
(Multiple Access protocol)

(5) Channel partiting protocal.

TDMA , FDMA , CDMA

⟹ TDMA :-

(Time Divisiom multiple Access.)



- we can not devide frequency, we only devide time.
- perticular time slot assign to statiom.

assign

T1 ⟶ statiom 1
T2 ⟶ statiom 2
T3 ⟶ statiom 3
T4 ⟶ statiom 4

- Time limit are assign to every slot and only data transmission are allow in this time.

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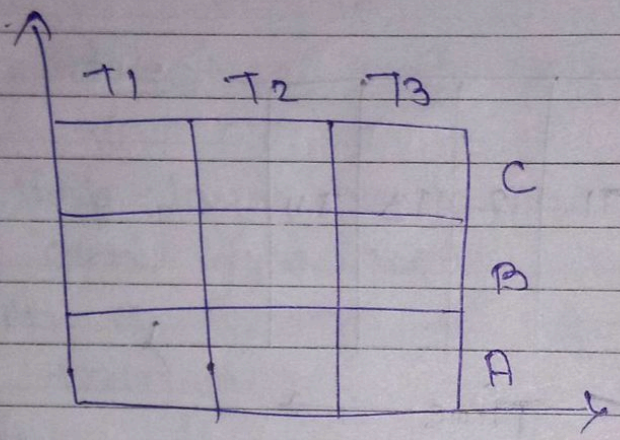
- use in radio system.
- Access to full length to frequency of channel.
- no groundband require for wide band system.



- efficency are increase.
- performance are increase and time decreace.

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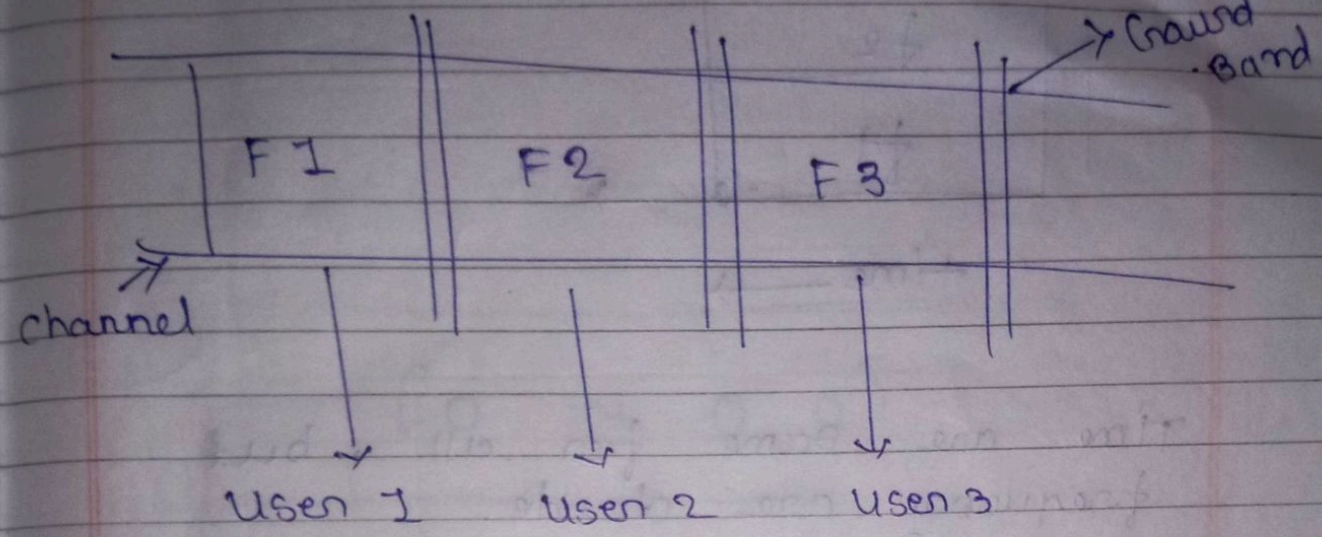
FDMA

(frequency division multiple Access.)



- in this we can devide frequency.
- Gaurd Band reduce infearance betweem two users.
- when multiple user access same channel at a time, them collisiom i's occur.
- then we can devide channel im to the frequency.

we can allocat

f1 to user 1

f2 to user 2

f3 to user 3

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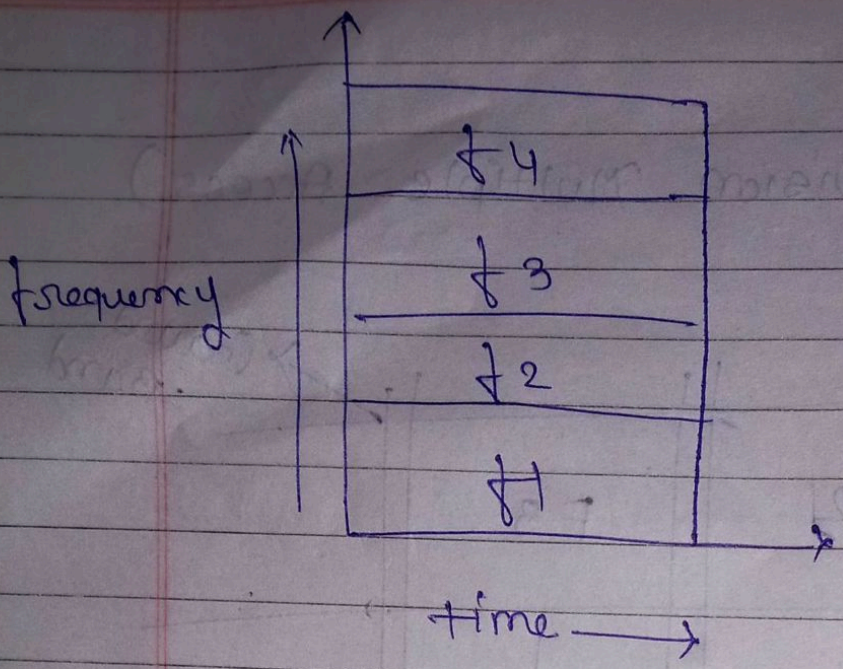

Time are same for all, but frequency are devide.

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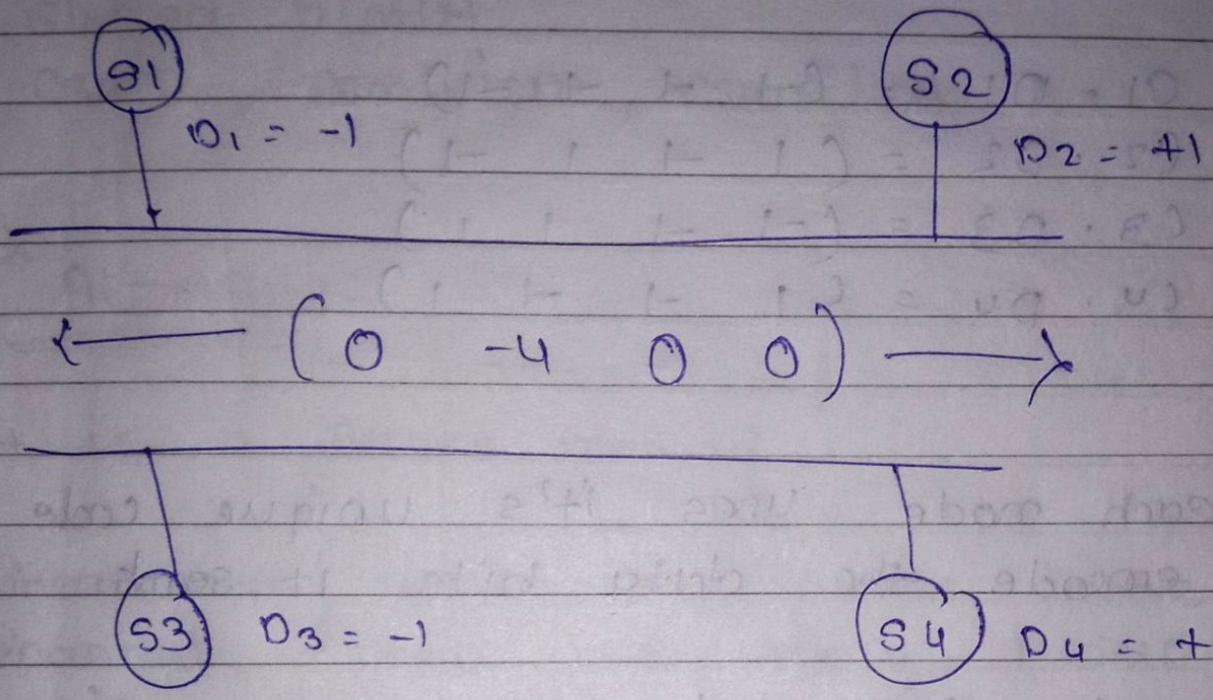
CDMA

(Code Division Multiple Access)

$C_1 = (1\ 1\ 1\ 1)$ $C_2 = (1\ -1\ 1\ -1)$



$C_3 = (1\ 1\ -1\ -1)$ $C_4 = (1\ -1\ -1\ 1)$

$C_1 \times C_2 = 0$

$= (1\ 1\ 1\ 1) \times (1\ -1\ 1\ -1)$

$= 0$

$C_1 \times C_1 = 4$

$= (1\ 1\ 1\ 1) \cdot (1\ 1\ 1\ 1)$

$= 4$

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Bit transfer

S1 ⟶ -1
S2 ⟶ +1
S3 ⟶ -1
S4 ⟶ +1

C1 · D1 = (-1 -1 -1 -1)
C2 · D2 = (1 -1 1 -1)
C3 · D3 = (-1 -1 1 1)
C4 · D4 = (1 -1 -1 1)

- each node uses it's unique code to encode the data bits it sends
- while TDM and FDM assign time slots and frequency, CDMA assign a different code to each node.
- if the codes are chosen carefully CDMA networks have the wonderful property.
- CDMA has been used in military systems for some time.
- now has widespread civilian use, particularly in cellular telephony.

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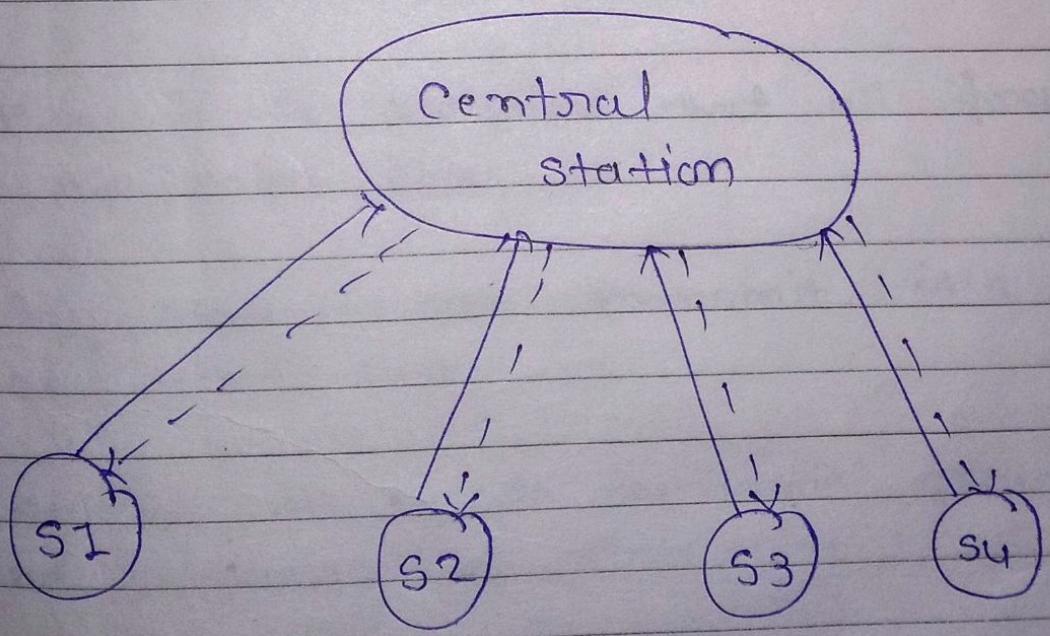
{6} Random Access protocols
(channel not Divided)

ALOHA
Slotted ALOHA
CSMA, CSMA/CD, CSMA/CA

ALOHA :-

- it is a Access method.
- when many users are access same channel then we apply ALOHA.
- also contention system.



- already decide time for transmit a frame.

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- frame length are fixed.
- we can start and send a frame then we can set back of time 0.
- send a frame and wait for Ack.
- if Ack are receive then it's success of transmission.
- otherwise aborted.

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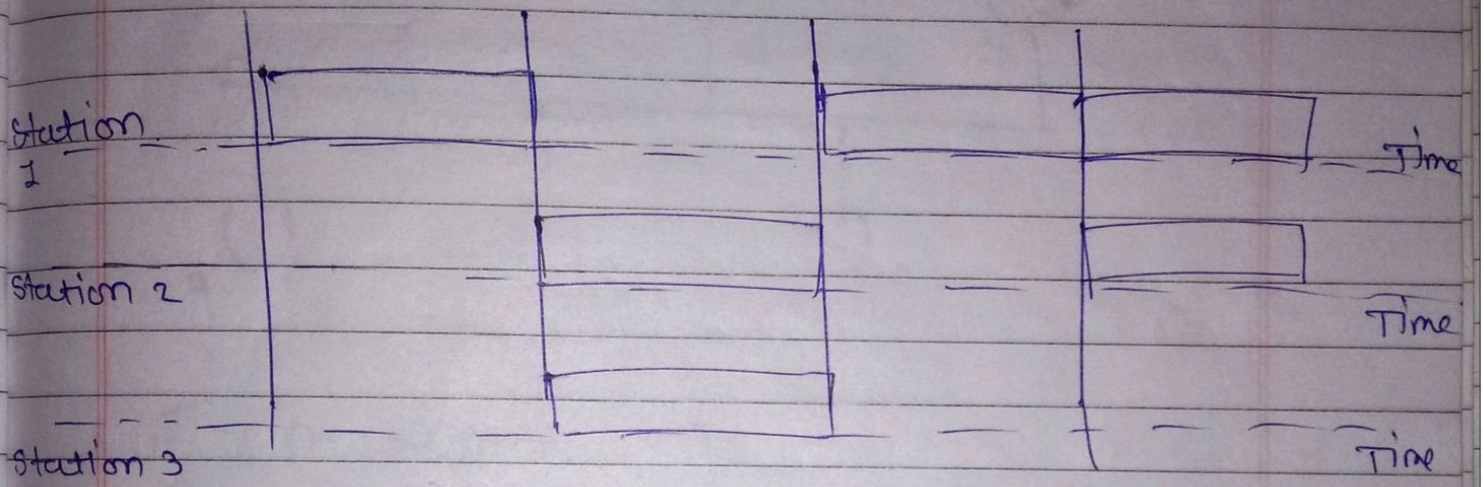
⟹ Slotted ALOHA :-

- it was developed just to improve the efficiency of pure aloha.
- in slotted aloha, the shared channel is divided into a fixed time slots.



- Here it is assumed that all frames are of same size.
- Any station can transmit it's data in any time slot.
- station starts to transmit frames only at beginning of time slot.

If the beginning of the slot is missed then station has to wait until the beginning of the next time slot.

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- If two on more station transmit in slot, all nodes detect collision.

- It makes the station to wait till the next time slot begine and allow each data frame to be transmitted in the new time slot.

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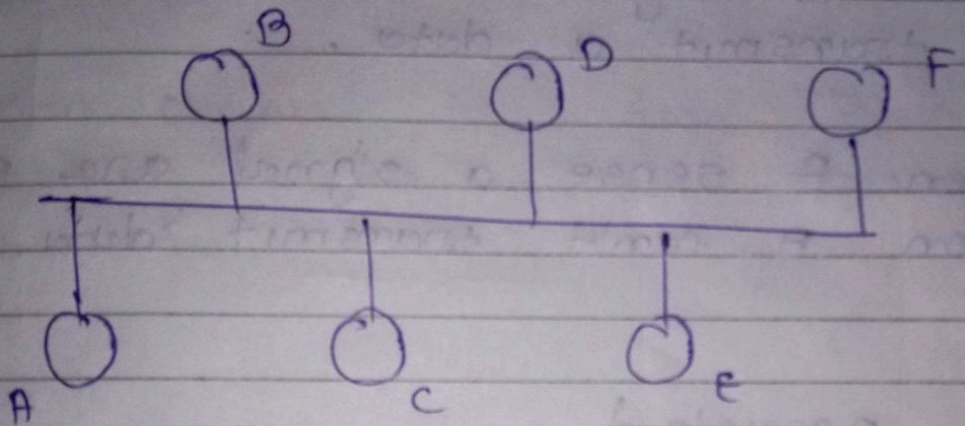
⟹ CSMA

(Carrier sense Multiple Access)

- To minimize the chance of collision
- increase the performance.
-



1 - persistent
0 - persistent
p - persistent
} variations.

- node start transmission, before transmission it sense the channel.
- we can not sense entire path.
- only check one by one node.
- for eg. A sense only A node.

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→ 1 - persistent.

- when A wants to transmit data,
- but first it check already signal are moved or not.
- when no signal are move, then A transmit a data.
- when A sense a signal are moved, then A can't transmit data.

→ 0 - persistent.

- when mediam are busy, they wait for random time of 0 amount.
- when time are finished A started transmission.

- for eg. ~~when~~ time are 1 hr., but task finished in 30 min.
- but A wait for 1 hr.

→ p - persistent.

↓

probility value

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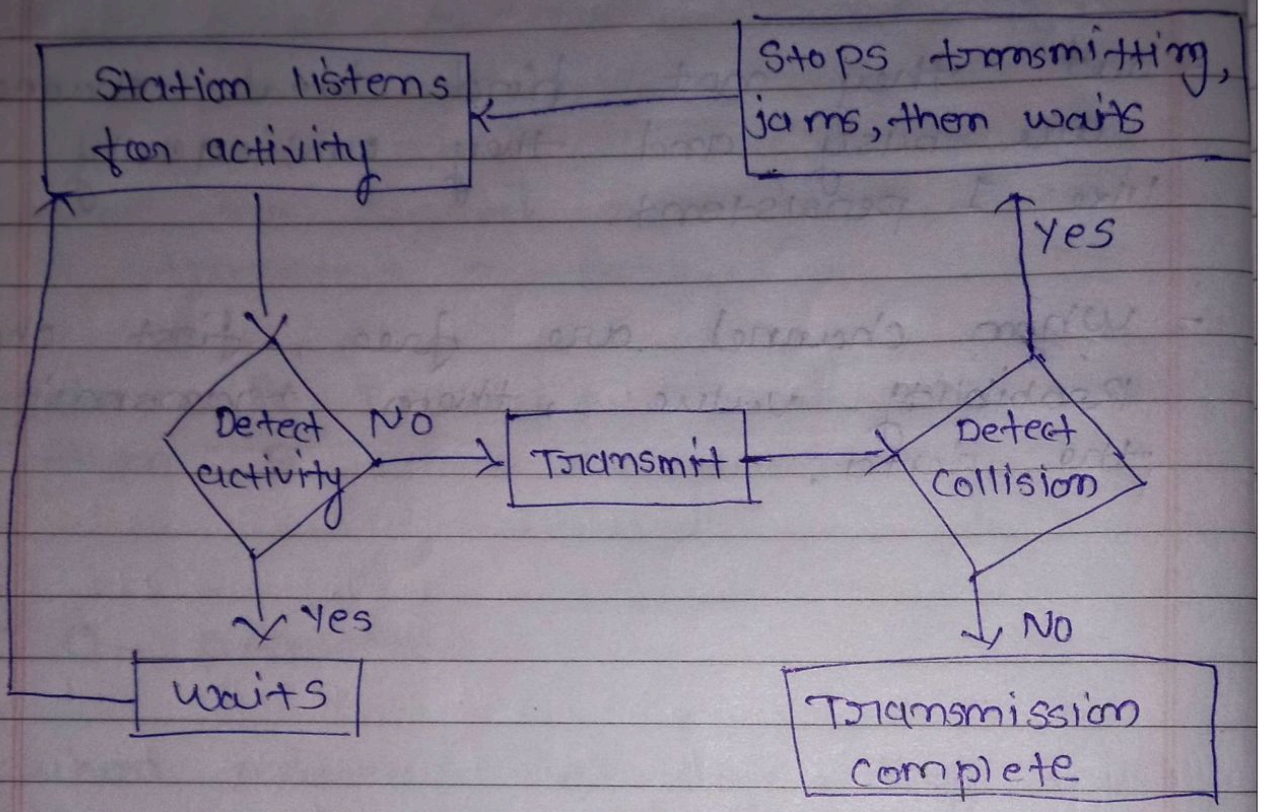
- it is mixture of 1 and 0 persistent.
- it checks continuasly when it find ideal, transmitt the data.
- but they not find ideal, channel are busy and they continuasly sense like 1 persistent.
- when channel are free, first check probiblity value, them transmit the data.

CSMA/CD

(Collision detection)



The algorithm of CSMA/CD is :

- When a frame is ready the transmitting station checks whether the channel is idle or busy.
- If the channel is busy, the station waits until the channel becomes idle.
- If the channel is idle, the station starts transmitting and continually monitors the channel to detect collision.

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- if a collision is detected, the station starts the collision resolution algorithm.
- The station resets the retransmission counters and completes frame transmission.
- if the maximum number of retransmission attempts is reached, then the station aborts transmission.
- otherwise, the station waits for a back off period which is generally a function of the number of collision and restart main algorithm.

(7). Describe the VLAN.

(virtual local Area Network.)

VLAN is collection of same type of:
- Devices
- Traffic
- Departments

VLAN are represented by a number which is known as VLAN ID.

each VLAN is assigned a different / unique ID.

in VLAN, the computers, servers and other network devices are logically connected to their physical location.

- improved security
- traffic management
- Make a network simpler

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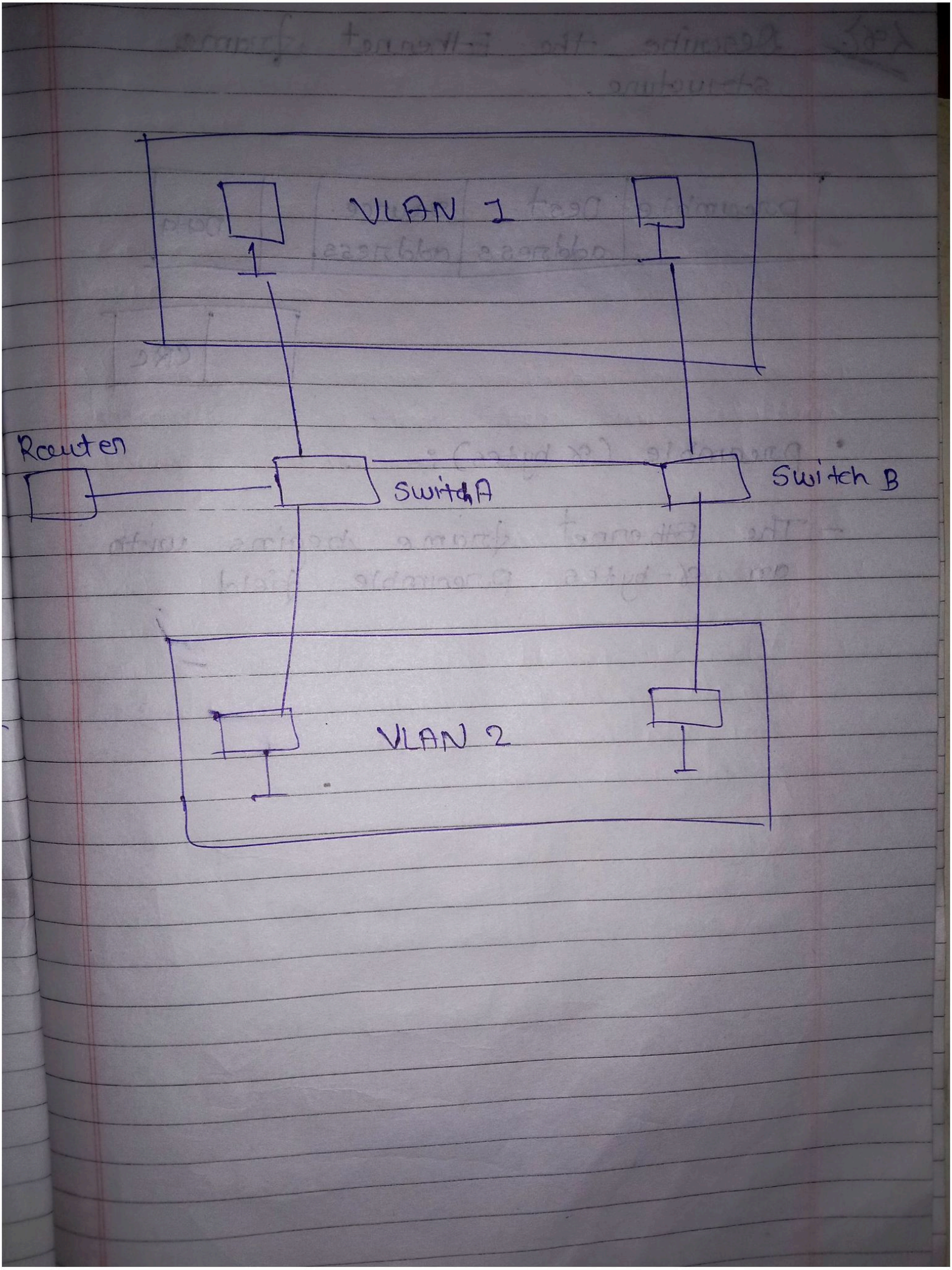

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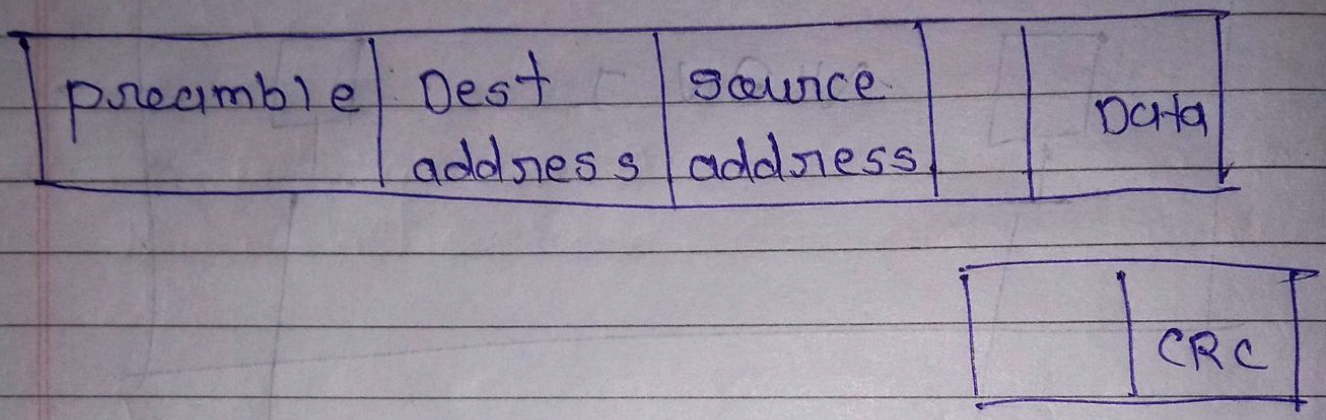
(8) Describe the Ethernet frame structure.



- preamble (8 bytes) :-

- The Ethernet frame begins with an 8-bytes preamble field.

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(9) Describe the polling and token passing.

⟹ polling :- token passing

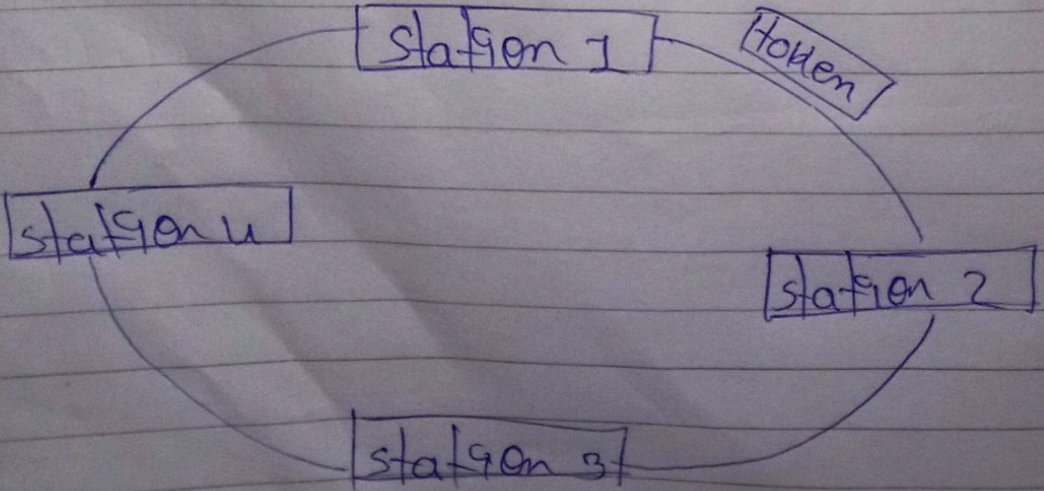
→ ~~In polling~~ In token passing scheme the station are connected to each other.

→ It is work link ring topology and structure are same as ring.

→ We can access the stations is using tokens.

→ token is special bit pattern or a small message, which circulate from one station to the next in some predefined order

→ token is passed from one station to another station in ring whereas increas of token bus, each station.



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node 1 might always send the token to node 2, node 2 might always send the token to node 3 and node 3 might always send the token to node 1.

when a node receives a token, it holds onto the token only if it has some frames to transmit.

otherwise, it immediately forwards the token to the next node.

If a node does have frames to transmit when it receives the token, it sends to a maximum number of frames and then forwards the token to the next node.

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<10>. Describe the bit stuffing and byte stuffing with suitable example.

⟹ Bit stuffing :-

data : 01110111 01110111

extra frame data : 00111100

send

00111100	01110111 01110111	00111100
00111100	01110011 10011110	00111100

receive

- we can send data and put extra bit in frame.
- extra bit are put in beginning and ending.
- And them calculate n-1 bit and put 0 aften n-1 bit in original datea.

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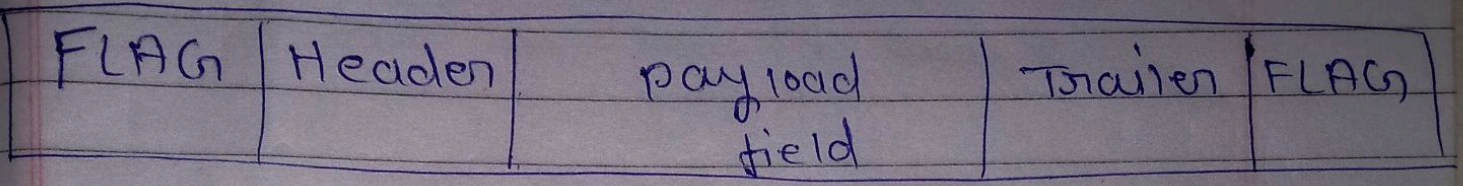
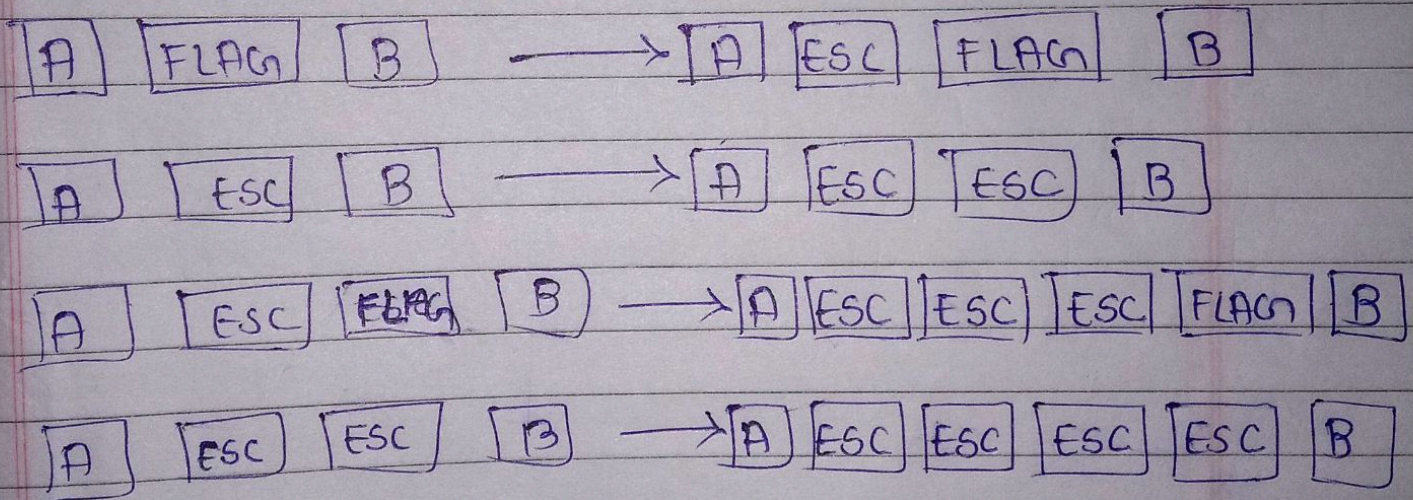
⟹ Byte stuffing :-



(a).

Original character → After stuffing



(b).

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